GISP Study Guide

Conceptual Foundations

101 Understanding of datums coordinate systems, and projections

Spatial Reference system

- Coordinate reference system
- Framework used to precisely measure locations on or near the surface of the earth

GEOID

- Shape that the surface of the oceans would take under the influence of gravitation and rotation alone in the absence of other influences such as wind and tides

- Equipotential surface, 3D surface where pole of gravity is a specified constant
- Measured and interpolates surface and not mathematically defined
- Used to reference heights in GIS

Spheroid

- Sphere-like but not perfectly spherical object
- Mathematically defined object

Geocentric

- Center on the earth's center of mass

Geodetic

- Related to geodesy or surveying

Ellipsoid

- Spheroid with a semi minor axis in polar direction and semi major axis in equatorial direction
- Equatorial radius is always greater than the polar radius in an earth ellipsoid

3 Common Spheroids

- Clarke 1866
- GRS 1980
    - Geodetic Reference System
    - Preferred spheroid globally
    - Used for NAD83 and ITRF (International Reference Frame)
- WGS 1984
    - World Geodetic System
    - Used for WGS 1984 reference frame

Datum

- Reference surface that provides known locations to begin surveys and create maps
- Parts of datum include ellipsoid, origin point, coordinate system, and set of control points that can be used for starting points for measurement
- Basis of geodetic survey work
- Use benchmarks from geodetic survey
- Horizontal and vertical datums
- Geocentric, or linked to a location on the earths surface
- Also known as reference frames or geodetic datums

Horizontal datum

- Horizontal data

102 Understanding of representation of discrete features and continuous phenomena in GIS

GIS Dataset

- Must have features, attributes, and spatial information

Discrete Data

- Only occurs at specific locations and has well defined boundaries
- Usually in vector format
- Also called categorical or discontinuous data

Continuous Data
- Each location is a measure of something
- Often in raster format
- Examples: elevation and temperature

Vector Data Model
- Is a coordinate based model that represents features with points, lines, and polygons
- Stores geospatial features as entities

Vector Advantages
- More efficient storage, because they dont require values to be stored for every location
- Accuracy is only limited by the measurement process
- Vector file can support topology and record relationships between features
- Able to be queried to find specific feature

Vector Disadvantages
- Are more compoles from a programmatic perspective
- Not as easy to explain how it works compared to raster
- Computationally intensive than raster
- Not good at variable comparisons

Point
- Coordinate location describe by x,y, and sometimes z
- X is longitude
- Y is latitude
- Z is elevation

Lines
- Is an ordered set of connected points
- Begin and end at nodes
- Vertices define shape between nodes

Polygon
- Set of vertices on a planar surface that begin and end at the same point
- Also known as bounded area

Intersection
- Point common to multiple lines or surfaces

Node
- Point at which a line feature end, intersect, or branch
- Are central or connecting points

Pseudonode
- Is node at the end of a line feature that connects a node of only one other feature
- Makes it appear as though two separate features are one
- Indicee where two feature may be merged

Dangling node
- Point at the end of a line with no connections

Edge
- Is a line that forms a boundary between two features

Curve
- Is a line that is defined mathematically rather than by a series of connected vertices

- May also be called parametric or bezier curve

Centroid

- Central location of a polygon
- Can be located outside of the polygon

Area

- Amount of space a polygon contains

Perimeter

- Is the total measure of the length of the outside of a polygon

Raster Data Model

- Is a representation of the world as a surface divided into a regular grid of cells
- Useful for storing data that varies continuously
- Can have single or multiple bands

Raster Bands

- Values per cell

Raster Advantages

- Easier to understand and explain to untrained person
- Good representing continuous highly variable surfaces such as terrain of earth
- Easy to input and output of computer programs
- Easier to perform map algebra

Raster Disadvantages

- Have larger file sizes
- Can be compressed to reduce file size but compression techniques are not efficient with variable data
- Compression only efficient when large areas have same values
- Reduction of file size could result in data loss

Raster Components

- Can contains discrete or continuous range of values
- Include remote sensed imagery, surfaces, scanned maps, and classified images

Discrete **Raster**

- Cells are limited to predetermined categorical values, usually integers

Continuous Rasters

- Cells have gradually changing values
- Can have floating point values

Pixel

- Is the smallest of resolvable information in a raster or picture
- Called cells in raster models
- Cells are connected to tables with indices
- Attributes or bands of data are stored in tables connected to cells with indices

Bit depth

- More bits means the dataset has a greater range of data that can represent finer gradations of color
- 8 bits allow 256 different values
- 16 bits allow 65000 different values
- Elevation rasters usually require 16 bits

- Higher bit depth allows greater detail but is a larger file size
- Lower bit depth allows lower range of data but reduces file size

Digital elevation model or DEM
- Is a data model used to process, store, analyze, and display elevation data
- Is traditionally a digital terrain model

Digital Terrain Model or DTM
- Type of dem that aims to represent an idealized land surface where surface objects such as building and trees have been digitally removed

Digital Surface Model or DSM
- Is a a type of dem that represents a maximum integration within a grid cell
- Records tips of building, trees and other objects

Mixed Pixel
- Condition where more than one category of object is represented in a grid cell
- Possible solution to this problem are ranking categories giving them wight, a fuzzy classification, assign mixed category, or reduce pixel cell size

Extent
- Area of distance in real space over which some geographic entity exists
- In cartography and GIS, is the size of the real space being represented

Resolution
- Degree of detail to which a phenomenon is detected or represented
- GIS data is stored and rendered as some degree of representation resolution
- In rasters, resolution is defined in pixel size
- In vectors, resolution is the smallest distance that can be detected between two features

103 Knowledge of earth geometry and its approximations

Equipotential Surface
- Surface along a pole of gravity is a specified constant
- Gravity is measured with gravimeters and satellites to determine location of a equipotential surface
- Not smooth because the pole of gravity fluctuates around the earth due to differing masses at different locations

Mean Sea Level
- Is determined by measuring oceans water level in coastal places using long term average readings from tide gauges

Ellipsoid
- Spheroid with a semi minor axis in polar direction and semi major axis in equatorial direction
- Based on mathematically equation
- Earth ellipsoid semi minor axis extend from north to south and is the center earth rotates around
- Earth ellipsoid semi major axis is the width of the earth at the equator

Reference Ellipsoid

- Mathematically define surface that approximates the geoid which the truer figure of the earth
- Approximation of the shape of the earth

Orthometric Height
- Height above a geoid

Ellipsoidal Height
- Height above an ellipsoid

Geoidal Height
- Distance between an ellipsoid and a geoid
- Positive is geoidal is above ellipsoid
- Negative is geoidal is below the ellipsoid
- Also known as geoidal separation

Dynamic Heights
- Equipotential surface above an ellipsoid
- Different orthometric heights because of differing gravity and mass above the geoid

Heights
- Important to understand how these different types are measured

Gravimeter
- Device that measure gravitational force

Plumb Bob
- Is a weight suspended by a string that indicates direction of gravity
- Often attached to survey equipment such as total stations

Grace Satellites
- Pair of satellites that measure earth's gravity by measuring their distance apart as they orbit earth
- Drift closer together in stronger gravity fields and farther apart in weaker gravity fields

Spheres -vs- **Ellipsoids**
- Spheres can be used for small scale maps but larger scale maps need spheroid or ellipsoid that approximate shape of earth

Earth
- An oblate spheroid
- Rotates around at its minor axis
- 13 miles wider than at the poles

104 Knowledge of basic geomatics and relationships to GIS

Geomatics
- Discipline concerned with collection distribution, storage, analysis and processing and presentation of geographic data or geographic information
- Includes cartography, mapping, geography, geophysics, surveying, remote sensing, photogrammetry, gps, and gis

Geodesy
- Precise measurement and understanding of earth's shape, gravity field, and rotation

Surveying

- Science of determining terrestrial 2D or 3D position of points and distances and angles between them

Remote sensing
- Collecting data from a distance with technology such as satellites and lidar

Cartography
- Art of creating maps and spatial representations

Geographic Information Systems
- Digital tools for analyzing and visualizing geographic data

Global navigation satellite system
- GNSS
- Satellites based position and navigation technology

Hydrography
- Science of mapping water bodies and their features

Photogrammetry
- Science of obtaining reliable information about physical objects and the environment through the process of recording, measuring, and interpreting photographic images and other remotely sensed data

Meades ranch
- Geographic Center of the united states
- Benchmark for creating reference grids
- Anchor point for clarke ellipsoid of 1866

National geodetic survey
- Defines, maintains, and provides access to the national spatial reference system
- Provides framework for positioning activities in the united states
- Administered by NOAA
- Maintains benchmarks for the high accuracy reference network (HARD)

High accuracy reference network (HARD)
- Consist of 16,000 survey stations across the US
- Created to upgrade NAD83 with high precision GPS to improve accuracy
- Also known as high precision geodetic network
- Use by state plane coordinate systems and maintained by the national geodetic survey (NGS)

First geodetic problem
-

Second geodetic problem
-

Great circle distance
- Length of an arc between two points belonging to a circumference passing the center of a sphere
- Used for measuring distances on a 3D earth surface

Geodesic distance
- Shortest distance between two points across a curved surface
- Used on unprojected maps or very large distances that span hundreds of miles

- Similar to, but not same, as the great circle distance as the arc doesn't not have to belong to an arc passing through a sphere

Loxodromic measure

- Line of constant bearing or azimuths
- Broken into series of loxodromes to simplify navigation
- Rum lines are an example of loxodromes

Euclidean plane

- Geometric space with 2 dimensions
- Geometric space in which two numbers are required to determine the position of a point

Cartesian plane

- Euclidean plane with a cartesian coordinate system

Cartesian coordinate system

- Specified point on a plane with a pair of numbers called coordinates which are assigned distances to the point from two fixed perpendicular lines

Planar measure

- Made on a flat surface
- Can only be performed on project maps

Euclidean distance

- Straight line distance between two points
- Calculated from cartesian coordinates using pythagorean theorem
- Use don projected maps or over short distances

Manhattan distance

- Distance between two points measured along allowable routes of travel
- Ex: Distance a car must travel between two points
- Also known as taxicab geometry

Quadrant bearing

- Bearing of a line is measured as an angle from the reference meridian towards east or west
- Ex: n45e or 45 degrees northeast

Azimuth

- Direction measured in degrees clockwise from north on a azimuth circle
- Ranges from 0 to 359.9 degrees

Geospatial Data Fundamentals

201 Understanding of spatial data models and their associated planar geometries

Spatial models

- Framework for expressing mechanism for geographic process and design analytical workflows to understand them

- Help solve problems make decision and understand how phenomenon change through space and time
- cartographic , simple spatial, spatio-temporal models are main 3 categories and are not mutually exclusive

Cartographic models
- Automate map analysis and processing
- Outputs are nominal or ordinal
- Solve problems via spatial layer combination, overlay, buffers,m reclassification interpolations, mapo algebra, and other
- Static in time
- Not temporarily dynamic but can be used to analyze change and does not create new temporal data but can compare points in time
- often create intermediate data layers not apart of final output
- Often require weighted criteria or layers
- often used for multi criteria decision making analysis and suitability
- Can be represented with flowchart

Multicriteria decision making analysis
- Method to evaluates multiple conflicting criteria to support decision making
- Combines and weighs multiple spatial variables into a spatial indicator
- Same as suitability analysis
- Cartographic model

Suitability analysis
- Ranks land for various purposes
- Twop ranks, discrete and continuous

Discrete ranks
- Occur when land area is divided into predetermined set of categorical values

Continuous ranks
- Varies along a scale

Flowchart
- Diagram of datasets, operations, and there sequence of use
- Modelbuilder most popular in GIS

Designing a cartographic model
- Define criteria, obtain data, develop flowchart or plan, assign ranking / weightings to criteria, execute model

Simple spatial models
- Apply equations to predict a specific continuous variable across space
- Outputs often interval or ratio data
- Outputs are often continuous
- Typically involve sample data and statistical fitting process
- Estimate important events, densities, or other characteristics like housing values, soil erosion rates or cancer frequency
- Study small areas
- Precursors to spatio-temporal models

Hydrologic models

- Type of simple spatial model
- Model waterflow
- Often use DEM and map algebra

Spatio-temporal models
- Dynamic in space and time
- Time causes changes in variables in model
- Represent processes
- Objects move across environment
- Use at least three dimension, two plane dimension and one time dimensions
- Can add z values for a fourth dimension
- Predict continuous fields or discrete objects

Deterministic model
- Produces same outputs when given same inputs
- Spatio-temporal model

Stochastic model
- Random generation that produces different outputs within given same inputs
- Spatio-temporal model

Process based model
- Involves process that cause changes
- Spatio-temporal model

Purely fit model
- Does not include the processes that cause changes
- Spatio-temporal model

Cell based model
- Set of functions driven by cell values that update cell values through time
- Spatio-temporal model

Cellular automata
- Operate on a single cell based layer using simple rules to derive next state from the current state
- Usually consists of color cells on a grid that evolves on the discrete time steps based on neighboring conditions
- Usually has two states, on or off, or black and white

Agent based model
- Agents moved across digital landscapes
- React to their environments based on a set if rules
- often used for traffic analysis

Layers
- Spatial datasets or coverages

Vector model
- Coordinate based data model that uses points lines and polygons to represent features

Raster model
- Rectangular arrays of regularly spaced squared grid cells, where each cell has a value or attribute

Topological model

- Feature need to be connected using specific rules
- Usually built within vector data models

Network model

- Collection of topological connected network elements such as edges, junctions, and turns

Surface models

- Specific planar geometries

Digital elevation models (DEM)

- Representation of bare earth without trees, buildings, or other features
- Stowed as vector or raster
- Used to be developed from topographic maps but are now from LIDAR point cloud files
- USGS develop them as same extents as 1:24000 topos
- USGS compiles into a single DEM for national dataset or NED
- USGS are ASCII plain text files, contain georeference info such as point data information of the surface of earth

Tessellation

- Covering of the surface, often a plan using one or more geometric shapes called tiles with no overlaps or gaps
- Raster, DEMs and TINs are tessellations
- Can also be used for graphic art

Triangulated irregular network (TIN)

- Portions vector data into contiguous non-overlapping triangles

TIN advantage

- More efficient with storage than DEMs and contour lines
- Provide high precision over variable terrain and efficient storage space over homogeneous terrain

TIN disadvantages

- Require very accurate data source
- Production is computer intensive

Delauney triangles

- Triangle between three points whose circum circles do not contain any points
- Used to create triangulated irregular networks

LANDIS

- Famous stochastic model that simulates forest growth
- Administered by US forest service

Relational data model

- Logical way of organizing data in tables with rows and columns to manage and retrive data
- SQL, data definition, and query language

Object oriented data model

- Data management structure that stores data as objects that have attributes
- Easy to understand and interpret
- Kep components include classes, methods, and inheritance
- Class is a collection of similar objects

- Method represent behavior of objects
- Inheritance means new classes can inherit attributes of base classes

Object relational data model
- Combines elements of relations and object oriented relationship models
- Examples include sql server, oracle, postgresql, and the geodatabase

Bolstad model theory
- Cartographic models are temporary static, combine spatial data sets, form operations and develop function for problem solving
- Spatio-temporal models model dynamics in space and time and time driven processes
- Network models model flow and accumulation of resources

Goodchild model theory
- Data models have entities and fields as conceptual models
- Static modeling transforms inputs into outputs using sets of tool anf functions
- Dynamic modeling is iterative, includes sets of initial conditions and applies transformation to obtain a series of predictions and time intervals

Demers model theory
- Models can have a descriptive purpose described as a passive model that describes the study area
- Active models determine best solution to a problem
- Models can have stochastic methodology and based on statistical probabilities and are deterministic
- Models are based on functional linkages and interactions
- Model can have inductive logic, general models are based on individual logic and are deductive
- Model go from general to specific using known factors and relationships

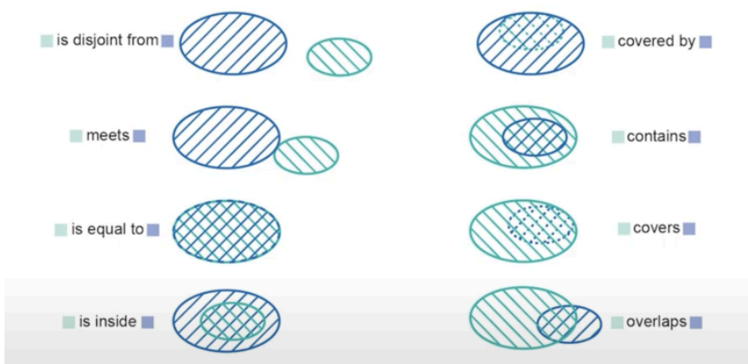
202 Understanding of spatial data relationships

Tolbers first law of geography
- States that everything is related to everything else but near things are more related than distant things
- Foundation concept for spatial data relationships

Topology
- How point line and polygon features share geometry
- Reduce file size by recording shared lines or points only once
- Usually consists rules of adjacency, connectivity and enclosure

Topological relationships
- Spatial relationships that characterize the relative position of spatial objects
- Include equals, disjointed, intersects, touches, contains, covers, covered by, within, crosses, and overlap
- Equals: two features are topologically equal
- Disjointed: when two features have no points in common, form a set of disconnected geometry
- Intersects: two features have some common interior points

- Touches: two feature have at least one boundary point in common but no interior points
- Contains: feature B is within feature A
- Covers: every point of feature B is a point of feature A
- Covered By: every point of feature A is a point of feature B
- Within: feature A is within feature B
- Crosses: feature A crosses feature B
- Overlaps: when feature A and feature B have common interior points



Topology rules
- Model spatial relationships between features or between feature classes
- Most popular is ArcGIS geodatabase

Polygon topology rules
- Must not overlap
    - Polygons must not overlap within a feature class
- Must not have gaps
    - Must not have a void between them within a class
- Contains point
    - Each polygon must contain a point within another feature class
- Must be covers by feature class of
    - Polygons must be covered by other polygons of another feature class
- Boundary must be covered by
    - Polygon boundaries must be covered by lines of another feature class
- Must not overlap with
    - Polygons must not overlap polygons of another feature class
- Must be covered by
    - Polygons must be covered by a single polygon from another feature class
- Area boundary must be covered by boundary of
    - Boundaries of polygons must be covered by the boundaries of polygons from another feature class
- Must clover each other
    - Polygons in two feature classes must covered each other

Line topology rules
- Must be larger than cluster tolerance
    - Vertices that fall within cluster tolerance are defined as coincident and snapped together

- Must not have pseudo nodes
    - End of a line cannot touch the end of only one other line
    - Used in hydrologic analysis
- Must not have dangles
    - End of a line must touch another line in its feature class or itself
- Must not self overlap
    - Lines cant overlap itself or have two common vertices
- Must not overlap
    - Lines cant overlap any line in their feature class or have two common vertices
- Must not self intersect
    - Lines cant cross or overlap themselves
- Must not intersect
    - Lines cant cross any other line in their feature class
- Must not intersect with
    - Lines must not cross any line in another feature class
- Must be single part
    - Lines must be one series of connected segments
- Must be covered by feature class of
    - Liens must be coincident to other lines in another feature class
- Must not intersect or touch interior
    - Lines can only touch at their ends and must not overlap each other within a feature class
    - Used for lot lines
- Must not intersect or touch interior with
    - Lines can only touch at their ends and must not overlap lines in another feature class
- Must be covered by boundary of
    - Lines must be covered by lines in another feature class
- Must not overlap with
    - Lines cannot overlap or have common connecting vertices to any part of a line within another feature class
- Endpoint must be covered by
    - Endpoints of lines must be covered by points in another feature class

Point topology rules

- Must coincide with
    - Points must be coincident with points in another feature class
- Must be disjoined
    - Points cannot overlap within the same feature class
- Must be covered by endpoint of
    - Points must be covered by the end points of lines in another feature class
- Must be covered by line
    - Points must be covered by lines from another feature class
- Must be properly inside polygon
    - Points must inside polygons of another feature class

- Must be covered by boundary of
    - Points must touch boundaries of polygons of another feature class

Adjacency
- Share a common boundary or touch each other

Contiguous
- Connected or share a border
- In a map, this represents features that are physically touching

Overlap
- Feature that occupy the same space

Proximity
- How close features are to each other

Spatial join
- Connect or join data based on their spatial relationship

Networks
- Interconnected set of points and lines that represent routes
- Also known as geometric networks and network datasets

Colocation analysis
- Examines local patterns of spatial association between two categories of point features
- Quantifies how often certain features occur together in proximity

203 Understanding of data quality

Data quality
- Overall suitability of a dataset

ISO data quality
- International standards organization states 5 types of data quality elements of a dataset are completeness, logical consistency, positional accuracy, temporal accuracy, and thematic accuracy

Completeness
- How well a dataset captures all the information it is intended to represent

Logical consistency
- How well data from different data sources are integrated

Positional accuracy
- How close the location correspond to the true values also known as geometric accuracy

Temporal accuracy
- How appropriate time frames in which data was measured for the use of the project

Thematic accuracy
- Is how different attribute values are to the true attribute values also known as attribute accuracy

Data lineage
- The sources, methods, timing, and persons responsible for the dataset

Necessary level of accuracy
- Depends on the needs of the customers

Fitness for use

- Is how well the data fulfills the needs of a project

Relevance
- Same as fitness for use

Remotely sensed data accuracy
- 4 types of accuracy are spatial, spectral, temporal, and radiometric

Spatial accuracy
- How geographically close data is to the true value

Spectral accuracy
- Refers to the parts of the electromagnetic spectrum that are recorded

Temporal accuracy
- Time or interval at which measurements are taken

Radiometric accuracy
- Refers to the amount of different values that can be recorded for measurement

Accuracy and precision
- Accuracy is how close the mapped values are to the real world values
- Inaccuracies occur in systematic bias or flaws in methodology
- Precision is how close measurement values are to each other
- Determined by scale at measurements were taken, consistency, randomness, and exactness of measurements (digits recorded after decimal place)

Timeliness
- Also known as temporal accuracy
- If data is collected at a point in time, it may not be accurate due to changes over time

Legal description
- Often determined by written description on deeds, GIS measurement may not line up with legal are drawn in legal deed or meets/bounds survey

Data validation
- Ensures accuracy and completeness
- Involves comparing measured data to real world true values
- Requires source of accurate data for comparison
- May involved developing accuracy statistics to compare standards

Data cleaning
- Involves removing errors and standardizing formats

Data sampling
- Samples of data can be analyzed to estimate quality of entire dataset
- Help with choosing locations, patterns, and number of samples

Sampling Methodologies
- 5 basic methods for sampling are systematic, random, cluster, stratified, and adaptive

Systematic sampling
- Whole study area is equally sampled
- May introduce bias in data if pattern in measured variable that coincides with sampling interval
- Not good for datasets with areas of high variation

Random sampling
- Involves selecting point locations using random numbers

- Reduces chances of bias because its unlikely to capture pattern within landscape
- Does nothing to distribute samples with areas of high variation

Cluster sampling

- Cluster locations are dialect by systematic or random process
- Cluster shape may be systematic or random
- Primary benefit is reduced travel time

Stratified sampling

- Partitions a population into subpopulations and takes samples from the subpopulations
- Reduces chances of subpopulation over or under represented compared to another subpopulation

Adaptive sampling

- Has higher sampling densities where the features are more variables
- Increases sampling efficiency because small scale variation is more heavily sampled
- Represent large homogenous area with less samples

Root mean square error

- Square root of the average of squared errors
- Indicates distribution of errors from the mean

Mean absolute error

- Average error
- Less sensitive to outliers than root mean square error
- Does not square distances from the mean

Confusion matrix

- Asses accuracy of image classification based on additional ground truths
- Used to produce a kappa index of agreements where 1 equals 100% accuracy and 0 represents 0% accuracy
- Also known as error matrix or error table

Error frequency histogram

- Summarizes spatial data error
- Smallest and largest errors, mean, and most commons areas are represented
- Bar chart of range frequencies

Error frequency threshold

- Value that is specified portion of the errors occurring above or below

Quality control

- Product based, and is reactive
- A corrective tool
- Involves a specific team that tests for defects
- Focuses on defect identification and correction after data collection

Quality assurance

- Process based and is proactive
- Managerial tool
- Involves the whole team
- Focuses of defect prevention before data collection
- Periodic audits are usually a component

Data accuracy

- Is how well a representation represents the true shape, location , and characteristics of a phenomena in GIS
- To determine accuracy requires known the true values
- May be reported as a percentage of data that is accurate, average error magnitude or percentage of points that are above error threshold

Precision and scale of numbers

- Is different precision and scale of a dataset

Number precision

- Precision of the number 123.123456 is 9 because there are 9 total digits

Number scale

- Scale of the number 123.123456 is 6 because that are six spaces after the decimal point

204 Understanding of data resolution

Resolution

- Image resolution is most common known type of resolution
- Images are raster data types
- Higher resolution has small cell size, higher DPI, records more detail, and causes larger file size
- Lower image resolution has large cell size, lower dpi, records less details, and causes smaller file size

Dots per inch

- The amount of cells that exists per inch in a raster or image resolution

Raster resolution

- Is the cell size of a raster
- Should be high enough to capture required detail but low enough for computer storage and analysis be performed efficiently
- Higher raster resolution causes higher accuracy, slower display speed, slower processing speed, and larger file size
- Lower raster resolution causes lower accuracy, faster display speed, faster processing speed, and smaller file size

Cell size

- Cell size should be considered based on spatial resolution of input data, application and analysis to be performed, size of result database compared to disk capacity, and desire response time

Raster resampling

- Resampling to larger cell size will reduce accuracy and file size
- Resampling to smaller cell size will increase file size and not increase accuracy, no inherit benefits
- Downsampling will reduce file size

Radiometric resolution

- Amount Of different measurements that a data band can record
- Equals 2 to the power of bits used to record a band of data
- 4 bits can store 16 different values

- 8 unsigned bits can record 255 different values
- 8 signed bits can record 128 to 127 values

Temporal resolution

- Frequency at which data is measured at the same location
- Once a week is higher temp[oral resolution than once a month

Spectral resolution

- Ability of a sensor to distinguish between wavelength intervals of the electromagnetic spectrum
- Most remotely sensed data consists of measurement of visible light or infrared interval of electromagnetic spectrum

Vector resolution

- Smallest difference between adjacent positions that can be recorded
- Determines minimum vector size that can be recorded

Resolution and scale

- Map should have appropriate resolution for the scale at which it is supposed to be viewed
- Higher resolution data should be used for large scale maps
- Lower resolution data should be used for smaller scale maps

Scale

- Amount of reduction between real world and graphic representation
- Usually expressed as ratio or an equivalent

Display scale

- Scale at which the map looks right

Data density

- How many features per area are stored

205 Understanding of data validation and uncertainty

Data validation

- Ensures data represents real world phenomenon they intend to describe
- Data is validated by comparing to ground observations or model generated data
- Ensures accuracy and completeness
- Involves comparing measured data to real world true values
- Requires source of accurate data for comparison
- May involve developing accuracy statistics to compare data standards
- Measures accuracy

Model builder validation

- Validation checks if all steps of model have been incorporated correctly and function properly

Data uncertainty

- Possible unknown difference between real world and spatial data that represents it
- When reporting measurement results you can can include uncertainty range
- Can be due to systematic or random error
- Many cause of uncertainty

Semantic uncertainty
- Uncertainties that arise from discrepancies in the meanings applied to spatial data often results in ambiguity and vagueness
- Semantics closely relates to syntax

Semantics
- Meaning of a word or phrase

Syntax
- The arrangement and structure of words

Ambiguity
- Occurs when imperfect indicators of phenomena are used instead of the phenomena themselves
- Usually caused by multiple terms referencing the same feature
- Caused by direct or indirect indicators

Direct indicators
- Have clear correspondence to the mapped phenomena
- Example: measuring temperature with a thermometer

Indirect indicators
- Are surrogate links with the phenomena of interest
- Example: estimating temperature using altitude

Vagueness
- Lack of distinction between boundaries
- Lack of distinction between objects, classes, and categories

Fuzzy classification
- Vague classification
- Technique for creating categories in a situation where it's not possible to assign categories to data with 100% certainty
- Converts semantic descriptions into a spatial representation using tools that classify variables using fuzzy membership classes rather than boolean classes
- Membership of data to classes can be assigned using continuous scale of membership from 0 to 1
- 0 assigned to data not apart of a set
- 1 assigned to data that are part of a set

Epsilon bands
- Area around a line that has a high probability of containing the true line segments
- Larger bands are associated with lower quality data

Error matrices
- Table used to determine accuracy of categorical data
- Dataset values are assigned to one axis and true values are assign in same order on perpendicular axis
- Features that match are tallied diagonally
- Also called error tables

Monte carlo simulation
- Model used to predict the probability of a variety of outcomes when the potential of random variables are present

- Runs multiple simulations and averages results

206 Understanding of metadata

Metadata

- Metadata records document the who, what, when, where, why, and how of a dat resource
- Cna include info that documents content, source,lineage,methods,developer, coordinate system, extent,structure,spatial accuracy,attributes, and responsible organization for spatial data
- Helps data get discover, accessed, and used appropriately
- Kep components are title, abstract, key words, spatial extent, temporal extent, data format, data source, accuracy and precision, projection information, and access constraints

Content standards for digital geospatial metadata (CSDGM)

- Usually aligned with international standards organization standards or ISO standards
- 10 types of metadata by CSDGm are identification and dataset description, data quality, spatial data organization, spatial reference coordinate system, entity and attribute information, distribution and options for obtaining dataset, currency of metadata and responsible party, citation information, time period information, and contact organization

International standards organization

- Effort to develop metadata standards
- FGDC recommends following ISO metadata standards
- Similar to CSDGM

207 Knowledge of temporal data

Temporal data

- Data that represent a state in time
- Attribute fields with time extents

Static map

- Shows one period of time

Dynamic map

- Allow a user to slide between different time periods to show changes

Multiple maps

- Show changes over time with multiple static maps with differing time periods

Time values

- Time at which data was recorded
- Can be stored in a data, string, or numeric field

208 Knowledge of spatial data standards, including ISO, FGDC, and OGC

Interoperability standards
- How spatial data are served between networks of hardware and software systems
- Facilitate sharing and access to GIS data

Analysis standards
- Ensure correct analysis are used and that produces best information possible

Professional and certification standards
- Establish educations, knowledge, or experience of GIS analysts and improves likelihood of technology being used appropriately

Geospatial competency model
- Identifies set of core and sector industry geospatial abilities
- Identifies 40 critical work functions
- Developed by employment and training administration (ETA), geotech center, and industry experts
- Include personal effectiveness competencies, academic competencies, workplace competencies, industry wide technical competence, industry standard competence, management competencies, and occupation specific requirements
- Geospatial abilities include positioning and data acquisition, analysis and modeling, software application and development, teamwork, creative thinking, planning and organizing, and problem solving and decision making, working with tools and technology, checking and examining and recording, business fundamentals, reading, writing, mathematics, geography, science and engineering, communication and speaking, critical and analysis thinking, basic computer skills, interpersonal skills, integrity, professionalism, initiative, dependability and reliability, and lifelong learning

Geographic information science and technology body of knowledge (GISNT)
- Comprehensive outline of the concepts and skills unique to geospatial realm
- First produced by university consortium for geographic science
- Published association of american geographers in 2006
- Basis for the geospatial competency model

Spatial data standards
- Methods for structuring, delivering, and delivering spatial referenced data
- Four categories: media standards, format standards, accuracy standards, and documentation standards
- Media standards are for physical form of which data is transferred
- Format standards are for the file components and structures
- Accuracy standards are for the quality and positional and attribute accuracy
- Documentation standards are for how spatial data are described

Data accuracy
- Accurate representation represents the true shape, location, and description of a phenomenon in a GIS
- Accuracy matt be reported as a percentage of data that is accurate or as an average error

Federal geographic data committee (FGDC)

- Organization of federal geospatial professional chaired by the US department of the interior
- Leads development and imp,emetnation and review of policies, practices, and standards related to geospatial data in the United States
- Provides oversight and direction for geospatial decision across the federal government
- Standards include addressing, metadata, content, data quality, soil codes, and wetland classifications
- Oversees the national spatial data infrastructure or NSDI
- National standard for spatial data accuracy is NSSDA was created by FGDC and being replaced by the ISO standards
- CSDGM or content standard for digital geographic metadata was created by the FGDC and being replaced by ISO standards

National spatial data infrastructure (NSDI)

- Build data infrastructure for improved public and private section applications of geospatial data and decision making
- Provides tool to access and share spatial data
- Administered by the federal geographic data committee

Content standard for digital geospatial metadata (CSDGM)

- Developed by FDGC
- Requires agencies to document all new geospatial data it collects or produces using the standard and make standardized documentation electronically accessible to the clearinghouse network
- Usually aligns with ISO standards
- CSDGM standrsa says metadata should include title, abstract, date, geographic extent, projection info, attribute definition and domain values, identification information, data quality information, spatial data organization information, spatial reference information, entity and attribute information, distribution information, and metadata reference information

National standard for spatial data accuracy (NSSDA)

- Defined by FGDC
- Specifies number and distribution of sample points when performing accuracy assessment and prescribed statical methods used to summarize and report positional error
- States that accuracy assessment should use 20-30 evenly distributed test points
- Sources of truth can include GNSS, field surveys, or digital orthophotographs
- Has standards for assessment of point accuracy
- Does not have standards for assessing accuracy of linear or area features
- Five steps to apply NSSA are identifying test points, identify true values, collect positional measurements, calculate positional error and summarize it in a standard accuracy static, and record the accuracy statistic in a standardized form

Federal bureau of the budget and national map accuracy standard

- publishes the National Map Accuracy Standards

- National Map Accuracy Standards are Larger than 1-20,000 scale maps must have 90% of the points be within 1 -30th of an inch of the true value, Smaller than 1-20,000 scale maps must have 90% of the points be within 1-50th of an inch of the true value. Elevation data must have 90% of the points be within 1 half of a contour interval of the true value
- An accuracy assessment must use at least 20 sample points

Open spatial consortium (OGC)

- an international organization of companies, government agencies, and universities that develops, approves, and maintains standards for sharing geographic data and maps over the web
- mission is to improve access to geospatial information and make location information FAIR. FAIR stands for Findable, Accessible, Interoperable, and Reusable
- a neutral forum where government, industry, and the public are able to access the open geospatial system

OGC and ISO simple features

-

OGC web based standards

-

Web feature service

- returns discrete features
- provides an interface allowing requests for geographical features across the web using platform independent calls
- provides interfaces for manipulating operations including creating features, deleting features, updating features, retrieval, and querying features based on spatial or non -spatial constraints

Web map service

- returns static maps
- a standard protocol for serving georeferenced map images over the internet
- provides two request types, get capabilities and get commands
- get capabilities returns information about image format, available layers, coordinate reference system, and bounding box. Get map returns a map image

Web coverage service

- returns multidimensional coverages in raster format that represent space -time varying phenomena
- shares raster datasets on the web in a format that can be used as input for analysis and modeling

Web map tile service

- a standard protocol for serving cached map or image services over the internet
- has get capabilities and get tile commands
- get capabilities requests metadata about the service. Get tile requests individual tile resources

Web processing service

- is an interface standard that provides rules for standardizing inputs and outputs for invoking geospatial processing services as a web service.

- an interface that facilitates the publishing of geospatial processes and the client's discovery of and binding to those processes
- has get capabilities, describe process, and execute commands
- Get capabilities returns metadata, Describe process returns metadata and description of the process, Execute returns the outputs of a process

International standards organization (ISO)
- maintains international standards for GIS methods, tools, and services for data management through the ISO Technical Committee 211, or ISO slash TC211
- charges fees for these standards
- are often required by public and defense sector entities

ISO 19115
- is a content standard that defines what should exist in metadata
- created by Technical Subcommittee 211 to harmonize the content standard for digital geographic metadata with other standards
- endorsed by the Federal Geographic Data Committee

ISO 19139
- contains XML schema implementation of the ISO metadata standards
- endorsed by the FGDC

ISO 19125
- contains simple feature standards

Mandatory fields for ISO datasets
- The mandatory fields for ISO datasets include title, date, geographic location, language, topic category, abstract, and metadata

American national standards institute (ANSI)
- is the U .S. member agency to the International Standards Organization, or ISO
- made the spatial data standard

Spatial data transfer standard
- a method for archiving and transferring spatial data
- specifies exchange constructs such as format, structure, and content for vector and raster data
- being replaced by geographic markup language
- Federal Geographic Data Committee has proposed to withdraw the Spatial Data Transfer Standard

Spatial data transfer standard accuracy
- states that raster data must be 95% accurate
- states that an accuracy assessment must use at least 20 checkpoints, so only one of the 20 test points can fail accuracy standards
- Spatial Data Transfer Standard states that an accuracy test must be performed against an independent source of higher accuracy

Infrastructure for spatial information in europe (INSPIRE)
- the organization that oversees spatial data standards in Europe

American society for photogrammetry and remote sensing (ASPRS)
- nicknamed itself the Imaging and Geospatial Information Society
- has standards for geospatial data accuracy

Cartography and Visualization

301 Understanding of graphic representation techniques and implications

Cartography
- the art and techniques of making maps
- Informative cartography requires identification of the who, what, where, and how

Specifying colors
- three ways to specify color are red, green, and blue or RGB, cyan, magenta, yellow or CMYK and Hue Saturation Value, or HSV

Red, green, blue (RGB)
- Red, Green, and Blue is the additive color system
- The zero of each color creates black
- max value of 255 of each color creates white
- combination of two primary colors creates a subtractive color

Red, green, blue, alpha (RGBA)
- a fourth value is used that indicates transparency
- A is for alpha, this value indicates transparency
- Zero is fully transparent

Opaque
- is a lack of transparency
- 100% opaque is not transparent at all. 0% opaque is fully transparent or invisible

Cyan, magenta, yellow (CMYK)
- is the subtractive color system
- The subtractive color system creates colors with the bases, cyan, magenta, yellow, and black
- The last value, K, is for black
- The absence of all bases is white light
- Subtractive colors absorb parts of the light spectrum, leaving the light light light light. leaving the rest of the spectrum behind. Mixing all subtractive colors creates black. The addition of black or K is necessary to create true black
- Cyan absorbs red, reflecting green and blue. Magenta absorbs green, reflecting red and blue. Yellow absorbs blue, reflecting green and red. Cyan and magenta absorb red and green to leave blue. Cyan and yellow absorb red and blue, leaving green
- Yellow and magenta absorb blue and green, leaving red. The Subtractive Color System is how color is printed on paper
-

Hue, saturations, value (HSV)
- additive color system
- Hue is the additive color base determined by amounts of primary colors
- Saturation is the intensity of color
- Less intensity creates a lighter, whiter color.

- Value is the dimension of lightness or darkness
- Lower value creates a darker color

Feature maps
- shows real -world features represented by cartographic objects
- used for navigation and reference maps
- also called reference maps

Thematic maps
- a map of a particular subject matter or theme
- usually visualizes properties of geographic features that are not visible, such as temperature, language, or population
- best to use an equal area projection for thematic map
- Equal area conic projections are used for the US, and equal area cylindrical projections are used for the world
- types include chloroplath, dasimetric, isoplath, dot density, multivariate, proportional symbol, and graduated symbol maps

Choropleth maps
- a thematic map that uses shading, colors, or patterns to show quantitative data about geographical areas
- Each color or pattern represents a range of values

Dasymetric map
- a type of chloropleth map
- ancillary information is used to model the internal distribution of a phenomenon
- is just a chloropleth map where the areas have been divided into more areas using another layer

Isopleth maps
- Iserythmic or isopleth maps are maps where lines of equal value are drawn called contour lines or ranges of similar values are filled with similar colors or patterns
- represent a continuous surface

Dot density maps
- show the distribution of a quantitative phenomena where values and locations are known
- Dots are placed where locations of variables are

Multivariate maps
- A multivariate display A multivariate display has more than two sets of data on a single map

Proportional symbol maps
- On a proportional symbol map, the size of the symbol corresponds to the magnitude of the map

Graduated symbol maps
- On a graduated symbol map, the size of a symbol represents a range of values

Color ramp
- a thematic set of colors used to represent variation, order, sequential values, or different categories on a map
- are usually divergent or sequential

Divergent color ramp

- shows when values are above or below a central value
- For example, Elevation above or below sea level

Sequential color map

- uses a sequence of colors to indicate which values are smaller or larger than others

Cartometric map

- a map that accurately represents the relative position of objects and may be used as a source of data. USGS topographic maps are well -known cardimetric maps. Web mapping

Web mapping

- uses the internet to generate and distribute spatial data and maps
- usually provides a service where users can choose what is shown on a map

302 Understanding of map design principles and essential map elements

Map

- is a model of the earth at a reduced scale.

Map Purpose

- Maps are categorized by their purpose
- purpose types include descriptive, exploratory, narrative, and rhetorical or prescriptive are considered a category as well

Descriptive Map

- has a broad purpose showing the features on the landscape as accurately as possible

Exploratory Map

- does not have a particular message

Descriptive and Exploratory Map

- descriptive and exploratory purpose types often overlap

Narrative Map

- has a specific message to convey

Rhetorical or prescriptive map

- attempts to convince map readers to have a specific opinion

Narrative and rhetorical/prescriptive map

- narrative and rhetorical or prescriptive purpose types often overlap

Map composition

- is how well the layers and themes of the map work together

Map layout elements

- include title, legend, north arrow, scale, sources, projection, author, date, neat line, and metadata

North Arrow

- A north arrow points in the direction of geographic north
- If there is more than one north direction on a map due to the projection, the north arrow should be left off, or multiple north arrows can be used to show the varying north orientations

Compass
- shows directions of north, south, east, and west

Map symbols
- Symbols represent features on a map
- should be used that are easy to understand

Map Scale
- is the relationship between the size of an object on a map and the size of the actual feature on the Earth's surface

Large scale map
- A large scale map covers small areas
- Features appear large on a large scale maps
- Large scale maps have a large scale ratio

Small scale map
- Small scale maps cover large areas
- Features appear small on a small
- Small scale maps have a small scale ratio

Scale factor and error
- Large scale maps generally have less geometric error than small scale maps

Types of scale
- three types of scales that can be displayed on a map, verbal, fractional or representational, and graphic or bar
- An example of a verbal scale is one inch represents one thousand feet. An example of a representative or fractional scale is one to one thousand
- graphic or bar scale consists of a bar marked off like a ruler with labels outlining distances between places

Absolute scale
- absolute scale begins at zero

Relative scale
- relative scale begins at a point defined by the author and can progress in both directions

Symbolization variables
- Symbols have characteristics
- Symbol characteristics include size, shape, orientation, pattern, hue, and value
- Symbols are used to represent two types of information

Quantitative information
- is best represented by the size and value of symbols

Qualitative information
- is best represented by the shape, pattern, hue, or orientation of symbols

Map layer
- is a mechanism to display a geographic data set

Map layer order
- GIS layers should be ordered from top down in this order. Points, lines, values, and polygons

Neatline
- is a frame around all the map elements

Graticule
- consists of lines that represent constant latitude and longitude
- can be straight or curved depending on the projection

Point
- is a unit of measurement for symbols

Planimetric element
- is a feature that is independent of elevation
- Planometric elements on maps are often roads, building footprints, rivers, and lakes

Planimetric map
- is a map that does not include relief or elevation data

Basic elements of map layout design
- Balance
    - the arrangement of elements within the overall composition and whether these elements appear stable or unsettled
    - Balance results from visual weight and visual direction
- Hierarchical organization
    - is the perceived importance of map elements
    - consists of separate meaningful characteristics to portray likeness, differences, and interrelationships
    - involves the visual separation of a map and two layers of information
- Figure ground
    - also known as visual hierarchy
    - Visual hierarchy is a separation of the figure in the foreground and the background
    - Objects that stand out are figures
    - The background information is the ground
    - Symbols should appear as figures, and context information should appear as ground
- Visual contrast
    - is the degree to which elements are distinguishable from one another
    - is how map features and page elements contrast with each other and their background
    - Objects that stand out from their background have high contrast and elements that blend together have low contrast
- Legibility
    - he concept of making symbols the appropriate sizes
- Typography
    - is the style, arrangement, and appearance of text, point size, line length, and typefaces
- Serif fonts
    - have small lines at the end of strokes
    - are often used for graphic features like mountains, oceans, and lakes
- Sans serif fonts
    - do not have extending features called serifs at the end of strokes

    - are often used in mapping for city names, legends, and roads
- Italicized
    - s a cursive font based on a stylized form of calligraphic handwriting
    - often used in mapping for labeling water features
- Kerning
    - adjusts the spacing between characters in proportion to the size of the individual characters
- Tracking
    - adjusts space uniformly between characters
- Deceptive mapping
    - is a map purposefully distorted for gain

303 Understanding of surface interpretation and representation

~~~this section is weird and only is a few minutes~~~

304 Understanding of 2D and 3D visualization

2D Map
- have only two dimensions, length and width

Planimetric Map
- is a map that has no relief data
- All features on a planometric map are 2D features

2.5D Grid Representation
- is a two -axis grid in which each grid cell or vertex has one and only one Z value

3D Mapping
- n a full 3D mapping environment, objects or images can have additional dimensions of depth, providing an appearance of volume and realistic representation

Elevation
- is the height or depth of a terrain element with respect to a vertical datum plane

Relief
- refers to changes in elevation

3D Mapping
- include vector datasets with z -values, rasters, triangulated irregular networks, multi -patch datasets, point cloud datasets, LIDAR data, and digital elevation models

Contour Map
- is a map with contour lines

Contour Lines
- are lines of equal height above or below sea level
- Contour lines run perpendicular to the slope
- lines of equal elevation on a map that run perpendicular to the slope
- Terrain characteristics can be interpreted from contour lines
~~~

- Slope can be determined from contour line spacing

Isoline

- are lines on a map that connect points of equal value
- ISO means equal
- Types of ISO lines include isobars, isobaths, isochrones, isopatches, and contour lines

Isobars

- are lines of equal atmospheric pressure

Isobaths

- are lines of equal water depth

Isochrones

- are lines of equal time distance from a specific location

Isopachs

- are lines of equal thickness over an area. Isopatches are used in the oil and gas industry

Index Contours

- bold or thicker lines on a topographic map that indicate elevation and help visualize contour lines

Contour interval

- is the difference in elevation between contour lines. Supplemental contours

Supplmental contours

- are placed in areas where elevation change is minimal to delineate small features that would be missing without them

Elevation data

- is recorded in triangulated irregular networks, profile plots, raster data sets, contour line data sets, and in digital elevation models

Extrusion

- is creating 3D shapes out of 2D shapes

Relief shading

- uses variations in shades of gray or dark and light contrasts that provide a 3D effect by assigning grayscale values representing surface reflectance of sunlight

Hillshading

- shows the topographical shape of hills and mountains using shading

Shaded relief map

- is a depiction of terrain brightness given a terrain surface and sun location
- is developed from a DEM and a model of light reflectance
- diffused light is scattered by the atmosphere and illuminates shaded areas
- brightness of a cell depends on the local incidence angle

Hypsometry

- is the measure of land elevation relative to sea level

Hypsometric color

- is coloring or tinting elevation intervals between contour lines
- also known as elevation tinting

Vertical exaggeration

- increases a feature's height to emphasize subtle changes or create a more dramatic effect

Triangulated irregular network
- represents a surface of connected triangles
- created from point cloud files by creating the Delaunay triangles
- are good for capturing linear features like roads and bridges
- are efficient at storing data for large homogeneous areas but are computer intensive to make

Breaklines
- occur where the shape of a surface abruptly changes in a tin

Wireframe model
- represents a 3D object with lines

Digital elevation models
- is a representation of the bare ground without trees or buildings or other objects
- used to be created from topographic maps
- used to be created from topographic maps

Multipatch datasets
- contain GIS objects that store collections of patches to represent 3D objects
- A patch is a 2D shape

Point clouds
- is a set of data points in a 3D coordinate system
- LIDAR data is initially stored in point cloud files

Light detection and ranging or LIDAR
- is a remote sensing method that uses light in the form of pulsed laser to measure ranges on Earth
- RAW data is recorded in vector point cloud format
- RAW data from LIDAR is often converted to raster form for use in GIS

Local incidence angle
- is the angle between the incoming light ray and the surface normal

Surface normal
- is a line perpendicular to the local surface
- Slope and surface direction define the surface normal

Solar zenith angle
- is the angle between vertical and the sun's location

Solar azimuth angle
- is the angle between north and the sun's position in a clockwise direction

Spot heights
- are exact points on a map with their elevation displayed beside them

Block diagram
- is an oblique view of terrain that also includes cutaway views of the subsurface

Swiss style maps
- is a style that combines various methods of terrain representation such as relief shading, contouring, rock drawing, and screen patterns in a matter consistent with the long established practices of Swiss cartographers that creates a three dimensional relief impression

National elevation dataset or NED

- produced mosaics of the best available elevation data for the United States
- The National Elevation Dataset was produced by the USGS
- was retired in 2015 and replaced by the 3DEP Dataset

National 3D elevation project

- is a program managed by the USGS to produce high -quality 3D topographic data
- It uses LiDAR data
- quality of the National 3D Elevation Project is QL2

USGS topographic maps

- were first published in 1884
- are produced every three years
- The scale of a USGS topographic map is 1 to 24 ,000
- USGS topographic maps portray natural and human -made features within the themes of elevation, hydrography, place names, boundaries, structures, and land cover
- are geo -referenced to NAD83 or WGS84 datums and projected to the UTM projection

3D models

- are highly detailed digital models of physical objects
- often developed for planning bridges, buildings, monuments, and stadiums
- can use remote sensing or be constructed by hand in 3D design software like SketchUp

Data Acquisition

401 Understanding of digitization and other manual data collection and conversion methods

Digitizing

- is the process by which coordinates from a map, image or other sources are converted into a digital format in GIS
- Map or image resolution and scale impacts the spatial accuracy of digitized data

Manual digitizing

- involves human guided coordinate capture from a map or image source

Scan digitizing

- Heads up digitizing is also known as scan digitizing

Heads up digitizing

- Heads up digitizing involves scanning paper documents into digital files

Automatic digitizing

- Automatic digitizing uses automated processes to convert raster data to vector data

Point mode digitizing

- In point mode digitizing, the operator must press a button to sample each point

Stream mode digitizing

- In stream mode digitizing, points are automatically sampled at a fixed time or distance interval

- Too short a collection interval will result in redundant points. Too long a collection interval will result in the loss of spatial data

Minimum distance digitizing
- is a variant of stream mode digitizing that only records a point if it is some minimum distance from the previously digitized point

Undershoots
- are nodes that fall short of the intended location

Overshoots
- are nodes that extend past the intended location

Snapping
- is an automated process that repositions digitized points within a specified range of a feature to be coincident with that feature

Spline functions
- are used to mathematically smooth lines between digitized points
- A spline is a set of positive points

Point thinning
- removes vertices to reduce processing time and file size

Weed distance
- Point thinning methods often use a weed distance to reduce the number of vertices while maintaining the line shape

Lang method
- is a point thinning technique
- uses a spanning line that connects two non -adjacent vertices in a line
- he thickness of the spanning line is the weed distance, any points that fall within the weed distance of the line are removed
- This process removes more points on lines that are not complex or straight than lines that are complex or that have many changes in direction

Skeletonizing
- reduces the width of scanned lines to a single pixel

Registration
- s the conversion of digitizer or other geospatial data to an earth -referenced coordinate system

Primary data
- is data collected for the first time for a GIS project
- data may include new digitization from remote sensing or aerial photographs.
- Primary data can include new GPS or survey measurements

Secondary data
- is data derived from other sources that were not collected for the specific project the data is being used in
- usually consists of data sets from government entities that weren't specifically prepared for the project at hand
- A scanned map is a secondary source because transferring the data is a secondary source. data between media formats reduces accuracy and detail.

Data collection

- A high quality data collection process should include these four steps,
- First plan the project
- Second test data collection in a sample area
- Third check the accuracy of the sampled data
- Fourth collect the rest of the data

Types of measurement

- **Physical measurement**
    - is measurement of the physical properties of the Earth or its inhabitants
- **Observations of behavior**
    - is measurement of observable actions or activities of individuals or groups
- **Archives**
    - are measurements from records that have been collected for other purposes
- **Explicit reports**
    - are measurements of beliefs people express about things
- **Computational modeling**
    - takes measurements from models as simplified representations of reality.

Qualitative data

- is descriptive
- consists of non -numerical or nominal values that have no quantitative meaning
- used to group common data for use in non-statistical analysis

Quantitative data

- consists of numerical values measured on at least an ordinal level
- can be counted
- can be used for statistical analysis

Fuzzy tolerance

- is the distance at which two points are considered the same

Cluster tolerance

- is the minimum tolerated distance between vertices and a topology
- Vertices within the cluster tolerance are snapped together

Common feature problem

- occurs when the same feature is mapped in different locations from different sources
- Possible solutions to the common feature problem are you could use the most accurate data source, you could weight the sources and use an average location, you could determine the location from other accurate data sources

Distance conversions

- a U .S. survey foot equals 0.304800609 meters
- an international foot equals 0.3048 meters
- the U .S. survey foot is longer than the international foot
- one mile equals 5,280 feet or 1.6 kilometers
- 1 kilometer equals 0.62 miles or 1,000 meters
- 1 meter equals 3.28 feet
- 1 meter equals 10 decimeters equals 100 centimeters equals 1,000 millimeters

Angle conversions

- normally recorded in degrees, but sometimes degree measurements are converted to radian measure for analysis

Radian Measure

- One radian equals 57.3 degrees
- 180 degrees equals 3.14 radians
- 360 degrees equals 6.2 radians
- Pi equals 3.14 radians.
- Pi equals 180 degrees

402 Knowledge of field data collection

Gnss and surveying

- Field data collection is mostly done using two technologies, GNSS and surveying

Global positioning system

- The first satellite positioning system, NAVSTAR, is called a GPS
- uses the WGS84 geographic coordinate system

Global navigation satellite system or GNSS

- is the generic name for all satellite positioning systems including NAVSTAR and others
- is used for tracking, navigation, field digitizing and surveying
- GNSS can calculate the coordinates for location by triangulating signals from three to four satellites
- 4 satellites are required to calculate an accurate location quickly

Gnss segments

- three main components, a satellite segment, a control segment, and a user segment
- The satellite segment consists of satellites orbiting Earth and transmitting locational signals.
- The control segment consists of tracking, communications, data gathering, integration, analysis, and control facilities
- The user segment consists of GNSS receivers

Navstar

- is the GNSS, or GPS, system of the United States

Glonass

- is the name of the Russian GNSS system

IRNSS

- is the GNSS of India

Galileo

- is the European Space Agency's GNSS

Compass/BEIDOU

- s the GNSS of China

Almanac

- is the data broadcast by satellites about the satellite's status and location

Range distance

- is the distance between two objects
- For GNSS, the range distance is the distance between the satellite in space and the receiver on the ground
- The range distance for a GNSS measurement is calculated by multiplying the speed of light by the travel time

Receiver

- is an electronic device that records and processes data from satellites to determine three-dimensional positions

Range pole

- is an extendable pole on which a GNSS antenna is mounted
- improves accuracy and reduces multipath signals

Laser rangefinder

- may be attached to GNSS receivers to improve field data collection
- are used in conjunction with an integrated electronic compass which measures angles

GNSS height

- GNSS typically measures height above an ellipsoid, not above a geoid
- Orthometric height, or height measurements above a geoid, can be calculated by referencing a nearby control point

GPS signals

- contain the location of the satellite and the time at which the signals were sent

Code division multiple access or CDMA

- is the technique of embedding signals within signals
- GNSS satellites transmit signals at the same frequencies
- Signals are separated into channels
- Many types of data from different sources can be sent on the same frequencies used by GNSS receivers
- is also used in cellular phone communications

Carrier phase signals

- are more accurate than coded phase signals but take longer to calculate and require more expensive receivers
- Carry coded signals

Coded signals

- are less accurate than carrier phase signals, but faster to use, and receivers that utilize coded signals cost less than receivers that utilize carrier phase signals
- are modulated from the carrier phase signals
- L1 and L2 bands carry most of the signals used by GNSS receivers
- The L1 band carries the CA and P codes
- The L2 band carries just the P code

C/A signals

- stands for Course Acquisition Code
- is transmitted by all satellites. It is comprised of an L1 carrier modulated by a CA code

CM and CL signals

- are modulated on the L2 carrier
- CM stands for Civil Medium

- CL stands for civil long

I5 and Q5 codes

- I5 stands for in phase
- Q5 stands for quadraphase
- the I5 and Q5 codes are modulated in the L5 carrier, they are independent but synchronized

Megahertz

- measures millions of cycles per second

Standard positioning service or SPS

- is a free GNSS service that uses the CA code to calculate positions
- The standards are for satellite signals, not for receivers
- requires signals to generate accuracy of 7 .8 meters or less

Precise positioning service or PPS

- is the GNSS service reserved for military use and authorized civilian users
- uses the P code, which is modulated over both L1 and L2 carriers

Consumer/recreational grade GPS

- has 10 meter accuracy and uses 1 frequency, the CA code from the L1 frequency

Survey grade GNSS

- can reliably produce 10 centimeter accuracy
- uses 2 frequencies, L1 and L2, to correct signal distortions caused by ionospheric delays

Military grade GNSS

- reliably produces 1 meter accuracy
- uses two frequencies, the L1 and the L2

GNSS errors

- are caused by errors in satellite position data, errors in the satellite atomic clock or receiver clock time, ionospheric delays in signal transmission, multipath signals and the satellite orientation

Improving GNSS

- by averaging multiple position fixes using dual frequency receivers that collect information on multiple GNSS signals simultaneously by setting a low signal -to -noise ratio, which will ignore signals with high noise, which reduces multipath signals by using differential correction, and by setting a low PDOP threshold

Signal to noise ratio

- compares the level of a desired signal to the level of background noise
- A ratio higher than 1 to 1 signifies more signal than noise
- A higher signal -to -noise ratio indicates a more reliable signal
- A signal strength threshold can be set on a receiver to eliminate multipath signals
- Lower signal strengths will increase the amount of measurements recorded, but increase errors

User range error

- is a calculation of ranging accuracy
- the fundamental performance parameter of GNSS that describes the total error that affects the distance measurements between a satellite and a user's receiver

- It's the square root of the sum of the squares of the individual biases that contribute to the error, including atmospheric delay, satellite almanac inaccuracies, receiver noise, and multi path signal frequency

User accuracy

- is how close the device's calculated position is from the truth
- expressed as a radius

Positional dilution of precision or PDOP

- describes how good satellite configuration is for collecting accurate points
- PDOP ranges from 1 to 20
- A lower PDOP is better, 5 or less is considered good
- improves if the satellites are spread more evenly through the sky
- PDOP signal strength threshold can be set on a receiver. A higher threshold will increase the number of fixes recorded, but lowers accuracy.

Differential positioning

- uses information from two receivers to remove range errors and improve accuracy

Base station

- is a stationary receiver at a known location

- is used to estimate range measurement errors and improve the accuracy of measurements

Real time differential correction

- requires a communications link between the base station and a roving receiver
- Corrections are calculated from the base station and applied in real -time

Real time kinematic or RTK

- is used for high accuracy GNSS location measurements
- uses a dual frequency station and a rover
- uses differential correction as well as other improvement techniques to achieve centimeter level accuracy for surveying
- no uniform definition for what constitutes an RTK system across the field of GIS
- just means a highly accurate GNSS measurement system

Moving base station

- is when the base station is on a moving vehicle

Networked transport of RTCM via Internet Protocol or NTRIP

- is a subscription service that provides corrective GNSS data through a wireless connection
- This method requires a constant Internet connection and a subscription to a local NTRIP service provider who will have an infrastructure of fixed base stations
- main advantage of NTRIP is that it uses a network of RTK base stations that are already in place and therefore has no range restriction

Satellite based augmentation system

- is a system that obtains high accuracy by receiving differential correction data from stations around the United States and sending that data to the receivers to correct measurements
- The most advanced satellite -based augmentation system in the United States is the Wide Area Augmentation System or WAAS

Virtual references station

- uses a network of reference stations that monitor GNSS signals and log corrections for reference
- Data from the virtual reference stations and roving receiver are fed to control centers that perform corrective calculations

Continuous operating reference station

- is a GNSS reference station minimally composed of a GNSS antenna, a GNSS receiver, data storage, and power
- are managed by the National Oceanic and Atmospheric Administration
- There are over 2 ,000 Continuous Operating Reference Stations
- They provide information for differential correction GPS systems

Post processing differential positioning

- can't be used for navigation

Precise point positioning or PPP

- is an alternative to differential correction
- performs high accuracy measurements by taking long observations over tens of minutes or hours

Surveying
- determines the locations of points by measuring distance and angles to other points

Triangulation surveying
- uses a network of interlocking triangles to determine positions at survey stations
- was used prior to satellite positioning
- utilizes the concept that the distance of a side of a triangle can be determined by knowing the distance of another side and the angles between the sides

Plane surveying
- Is surveying on a planar or flat surface

Coordinate geometry or COGO
- is used to draw points, lines, and polygons using survey points and measurements
- consist of a starting point called a station and a list of directions called bearings

Traverse
- is a series of connected lines that start at a control point
- the basic operation performed in coordinate geometry

Benchmarks
- are known points across the United States with exact locations
- usually consist of a brass disc embedded in rock or concrete
- are maintained by the National Geodetic Survey of the National Oceanic and Atmospheric Administration
- often used as control points

Azimuth angle
- is measured in a clockwise direction from 0 to 360 degrees relative to grid north

Bearing
- specifies angles from 0 to 90 degrees by stating N or S for north or south, a measurement of degrees, and E or W for east or west

Transit
- is a survey instrument used for measuring horizontal angles
- It has a telescope that can flip over for back sighting and doubling of angles for error reduction

Theodolite
- is a survey instrument for measuring horizontal and vertical angles

Total stations
- is a survey instrument that combines the functionality of a theodolite with an electronic distance measurement device
- It emits laser pulses to an accompanying prism to measure angles and distance.

Electronic distance measurement or EDM
- is a method of determining the length between two points using electromagnetic waves or lasers
- is commonly performed by total stations

Angle of deflection
- is the angle between a line and the prolongation of the proceeding line
- It is recorded to the right or left of the proceeding line

Building information modeling or BIM

- is the generation and management of digital representations of the physical and functional characteristics of buildings and other physical assets

Field collection process

- The field collection process should include six steps
- First, determine what needs to be collected, inspected, or surveyed
- Second, determine how it will be collected
- Third, perform sample collections
- Fourth, review sample collections and adjust methodology
- Fifth, plan location and timing
- Sixth, start collection

Public land survey system or PLSS

- s a standardized method for designating and describing land parcels in the U.S.
- is not a coordinate system, but is often used as reference points for parcel surveys

Metes and bounds surveying

- was the most common way of surveying before the public land survey system
- describes parcels relative to features on the landscape
- Those descriptions are supplemented with angle or distance measurements.

403 Knowledge of automated data collection and conversion methods

Feature extraction

- refers to the process of transforming raw data such as satellite imagery, lidar point clouds or other geospatial data into meaningful features
- s performed for land cover classification, object detection, change detection and terrain modeling

Manual feature extraction

- human analysts manually identify and delineate features of interest
- highly accurate but time consuming

Automated feature extraction

- algorithms automatically detect and extract features
- uses deep learning to recognize patterns

Data scraping

- involves extracting data from sources that are not intended to be sources of data, such as websites
- efficient for large -scale data extraction

Remote sensing

- is the process of employing an instrument to acquire information about an object or phenomenon without making physical contact with it

Extract, transform, load

- is a data integration process used to combine data from multiple sources into a consistent format for loading into a data warehouse, data lake, or other target system

Extract

- raw data is copied or exported from various source locations to a staging area

Transform

- raw data undergoes data processing in a staging area
- The data processing can include cleaning, sanitizing, aggregating, and enriching data

Data cleaning

- removes inconsistencies, errors, and duplicates

Data sanitizing

- ensures data quality by standardizing formats

Aggregating

- combines data from different sources, as data enriching adds additional information to the data

Enriching

- adds additional information to the data

Load

- cleaned and transformed data is loaded into a target database
- is organized and structured for efficient querying and reporting
- can happen incrementally or in batch mode

Application program interfaces

- are sources for specific data from online databases
- APIs are structured and reliable
- Programs can link to APIs and automatically update as data from the API changes

404 Knowledge of remotely sensed data sources and collection methods

Remote sensing

- is the process of employing an instrument to acquire information about an object or phenomenon without making physical contact with the phenomenon
- Remotely sensed images can provide a large area coverage, extended spectral range, geometric accuracy and a permanent record
- Large area coverage captures data over large areas for low cost
- Aerial images are useful for measuring locations of objects, categorizing surface features, and used for a backdrop for other features

Remotely sensed images

- Can provide a large area coverage, extended spectral range, geometric accuracy, and a permanent record
- Large area coverage captures data over large areas for low cost
- Extended spectral range, scanner can detect wavelengths outside of eyesight
- Images can be converted to accurate spatial data

Aerial images

- Are taken from aircrafts from cameras
- Useful for measuring location of objects, categorizing surface features, and used for backdrop of other features

Aerial image of distortion

- Sources of situation in aerial images are terrain variation, camera tilt, film deformation, camera lens shape, sensor defects, and atmospheric bending

Relief displacement

- is the radial displacement of objects that are at different elevations due to terrain variation
- Higher elevations are displaced outward, lower elevations are displaced inward

Perspective distortion

- is caused by Camera Tilt
- Camera Tilt causes objects further away to appear to be closer together than equivalently spaced objects that are nearer the observer
- Tilt Distortion is zero in a vertical photograph

Radial lens displacement

- is distortion caused by a camera lens
- A radial displacement curve is developed for a mapping camera lens and used to correct radial displacement

Perspective view

- is not orthographic
- It gives a geometrically distorted image of Earth's surface
- In a perspective view, the relative position of objects is not accurate

Orthographic images

- are geometrically precise
- Geometric distortion has been removed in orthographic images
- Orthographic images are called orthophotographs
- Orthophotographs have uniform scale and true geometry

Orthorectification

- is the process of removing geometric distortions from images
- Images have tilt and relief distortions that make it difficult to accurately compare a map and an image of the same geographic area
- Once an image is ortho -rectified, features on the image are shown in their planometric locations

Photogrammetry

- is the science of making measurements from pictures, aerial photographs, and other remote sensing data
- Measurements are made from overlapping pairs of photographs called stereo pairs

Stereophotographic coverage

- is created when sequential photographs in a flight line overlap called end lap or adjacent flight lines overlap called side lap
- creates stereo pairs for photogrammetry

Stereomodel

- is a three -dimensional perception of terrain or other objects from viewing a stereo pair
- Stereo models can be used to measure terrain heights

Stereoplotter

- uses stereo photographs to determine elevations

Satellite images

- are recorded with satellite scanners
- Satellite images have a very high perspective, which significantly reduces terrain cost distortion
- Satellite images cover much larger areas than aerial photography and provide much less detail
- Most satellites are in polar orbits, which makes their images overlap less at the equator and more towards the poles
- Satellites are used in GIS to create land cover layers and monitor change.

Pixel size

- he pixel size of an aerial or satellite photography should be half the size of the smallest object you're mapping

Electromagnetic spectrum

- is the range of wavelengths of light energy
- Visible light is measured in micrometers
- The color blue is from 0 .4 to 0 .5 micrometers. The color green is from 0 .5 to 0 .6 micrometers. The color red is from 0 .7 to 0 .8 micrometers
- Infrared is used to measure vegetation coverage

Passive sensing

- senses radiation emitted from objects

Active sensing

- emits energy and measures the amount of energy returned
- radar and lidar are types of active sensing

Remote sensing resolution

- Types of remote sensing resolution include spectral resolution, spatial resolution, temporal resolution, and radiometric resolution.
- Spectral resolution is the ability of a sensor to collect and record energy in a defined wavelength interval of the electromagnetic spectrum
- Spatial resolution is the size of the area on the ground covered by one pixel
- Temporal resolution is how often data is being collected
- Satellite temporal resolution is based on their orbit
- Drin's temporal frequency is based on their flight frequency.
- Radiometric resolution is the sensitivity of a sensor to discriminate small differences in emitted or reflected energy for a pixel
- Radiometric resolution is determined by the bit depth of the sensor
- Greater bit depth provides a higher resolution

Single band raster

- has one value for each cell
- Scanned maps are single band rasters

Multi band raster

- have several values for each cell
- can be sourced from satellite images and aerial photography

Composite image

- is made up of the overlay of multiple bands

Spectral signature

- is how light reflects from an object
- are used to classify or categorize data

Satellite orbits

- Most satellites orbit the Earth twice a day at about 12 ,000 miles

Unmanned aerial vehicle

- are also known as drones or unmanned aerial systems
- UAVs are low -cost, easily deployed, and capable of producing large-scale, high-resolution photography

Recreational drone pilots

- can fly to 400 feet

Class G airspace

- Class G Airspace Class G airspace is from the surface to 1200 feet or less
- Class G airspace is uncontrolled airspace

Part 107 license

- is the remote pilot license

Drone weight

- 55 pounds or less

Drone flying rules

- Drones must fly within line of sight fly during the daylight of simple twilight fly at no more than 100 miles an hour drones cannot fly over people drones cannot fly from a moving vehicle in a populated area and drones must yield to other aircraft

LIDAR

- light detection and ranging
- is a remote sensing method that emits light in the form of pulsed laser to measure ranges to earth
- measurements are used to generate precise, three -dimensional information about the shape of the earth and surface characteristics
- A LIDAR instrument consists of a laser, scanner, and a GPS receiver
- does not penetrate clouds, smoke, or haze well
- Lidar data is usually initially stored in LAS point cloud files

LIDAR quality levels

- quality levels include QL0, QL1, QL2, QL3, QL4, and QL5. Zero is best, five is worst
- quality level is determined by the pulse spacing and vertical positional accuracy

Discrete return lidar

- collects a few returns, typically four
- Classification codes for LiDAR data are defined by the American Society for Photogrammetry and Remote Sensing, or ASPRS

First return lidar

- The first signal returned from a laser pulse will show the first objects the signal encounters, such as tree tops and branches

Last return lidar

- The last return from a laser pulse will show ground shape

Waveform lidar

- collects the full profile of a return signal

LAS format

- The American Society of Photogrammetry and Remote Sensing defines the standard LAS format for LiDAR data

Water in lidar

- Much of the information gained from LiDAR sensing is due to the laser's interactions with water
- Water absorbs infrared light
- Water shows up as red on infrared-sensed images
- 1 indicates healthy vegetation
- Negative 1 indicates dead plants or inanimate objects.

Normalized difference generation index or NDVI

- is used to assess plant health
- Healthy vegetation will reflect infrared pulses
- Values range from negative 1 to 1

Sources of lidar data

- **Orthoimagery**
    - is corrected to be an exactly top -down view
    - is aerial photographs or satellite imagery that is geometrically corrected or ortho rectified
    - Used to measure true distances because distortion has been removed due to topographic relief, lens distortion, and camera tilt
- **Oblique imagery**
    - is flown at an angle
    - shows the sides of buildings and features
    - can be used to measure the heights of buildings and features
- **Radar or Radio detection and ranging**
    - emits a beam of energy through an antenna and records the reflected energy
    - uses one wavelength and produces a monochromatic or one color image
    - penetrates clouds and inclement weather because water does not absorb its long wavelengths
- **Digital orthophoto quarter quad or DOQQ**
    - is a computer generated image of an aerial photograph in which the image displacement from terrain relief and camera tilt has been removed
    -

Sources of orthophotos

- **National aerial photography program or NAPP**
    - is a USGS program that provides black and white aerial photography and infrared imagery taken between 1987 and 2004
- **National agriculture imagery program or NAIP**
    - is a program from the USDA that provides ortho -photography and geospatial data during the growing seasons
- **National digital orthophoto program or NDOP**

    - is a consortium of federal agencies with the purpose of developing and maintaining national ortho imagery coverage in the public domain by establishing partnerships with federal, state, local, tribal, and private organizations
- **United states forest service or USFS**
    - takes aerial images to classify forest cover
- **National Aeronautics and Space Administration or NASA**
    - provides a diverse group of spatial datasets, including orthophotos, elevation, land use, ecosystem types, and a number of measures of vegetation productivity, phenology, structure, and health

405 Knowledge of acquisition, use, and limitations of crowdsourced and open source data and services

Crowdsourced data
- consists of information, opinions or work that is collected from a large group of people
- is typically sourced via the internet, social media platforms and smartphone apps

Advantages of crowdsourced data
- include cost savings, skill diversity and real time data

Disadvantages of crowdsourced data
- include reduced quality and accuracy, bias and representativeness, privacy and security, motivation and incentives, task complexity, lack of content, and cost and time

Crowdsourced sources
- **Waze**
    - is a subsidiary company of Google that provides satellite navigation software on smartphones that utilize GPS
- **Localwiki**
    - is a grassroots effort to collect, share, and open the world's local knowledge
- **Openstreetmap**
    - is a free, open geographic database updated and maintained by a community of volunteers via open collaboration

Open data sources
- **Natural earth data**
    - is a public domain map dataset featuring tightly integrated vector and raster data
- **USGS earth explorer**
    - is your one -stop shop for obtaining geospatial datasets from extensive collections
    - Users can navigate via an interactive map or text search to obtain Landsat satellite imagery, radar data, UAS data, digital line graphs, digital elevation model data, aerial photos, Sentinel satellite data, some commercial satellite imagery, including including ICONOS and or Vue 3, Land Cover data, Digital Map data from the National Map, and many other data sets
- **Arcgis open data hub**
    - have access to data that all other ArcGIS Online users make available to the public

- **SEDAC**
  - has data sets related to population, sustainability, and geospatial data
  - Datasets are related to human population distribution, human settlements and infrastructure, ecosystems, agriculture, wetlands, environmental treaty status, environmental sustainability, natural hazards, poverty, and pollution
- **Open topography**
  - facilitates efficient access to topography data, tools, and resources to advance our understanding of the Earth's surface, vegetation, and human -built environment
  - s based in the San Diego Supercomputer Center at the University of California San Diego and is operated in collaboration with colleagues in the School of Earth and Space Exploration at Arizona State University and the Earthscoped Consortium which is formerly UNavco
  - Core operational support for OpenTopography comes from the Division of Earth Sciences at the National Science Foundation
- **United nations environmental program or UNEP**
  - is the leading global authority on the environment
  - offers more than 15 ,000 items, from real -time data tools and platforms to key reports, publications, and more. fact sheets, interactivities, and more
- **NASA earth observations or NEO**
  - provide global data sets related to climate and environmental changes
- **Sentinel data hub**
  - makes satellite data, including sentinels and Landsat satellite data, accessible to be browsed or analyzed
- **Terra populus**
  - integrates data about population and environment from censuses, surveys, land cover data, temperature, precipitation, climate, and land use data sets
- **Geonetwork**
  - is a portal of free GIS data from the United Nations
- **ISGCGM**
  - is a free source of global data including boundaries, drainage, transportation, population centers, elevation, land cover, land use, and vegetation

Global spatial dataset infrastructure or GSCI

- attempts to coordinate collection and processing of data sets for global level analysis by the USGS

Shuttle radar topography mission

- is an international research effort that obtained digital elevation models on a near global scale
- Elevation models are arranged into tiles, each covering one degree of latitude and one degree of latitude and degree of longitude
- hey are named according to their southwestern corners

Advanced Spaceborne Thermal Emission and Reflection Radiometer or ASTER

- admits high resolution 15 to 90 square meters per pixel, images of the Earth, and 14 different wavelengths of the electromagnetic spectrum, ranging from visible to thermal infrared light
- Scientists use ASTER data to create detailed maps of land surface, temperature, emissivity, reflectance, and elevation. The National Geospatial Data Clearinghouse

National geospatial data clearinghouse

- is a distributed system of agency servers located on the Internet that contain field -level descriptions of available and planned digital spatial data, applications, and services
- This descriptive information, known as metadata, is collected in a standard format to facilitate query and consistent presentation across multiple participating sites.
- Clearinghouse uses standards -based web technology for the publication and discovery of available geospatial resources through the Geospatial Platform Portal
- provides data for the U .S. USGS Orthorectified Aerial Photographs

USGS orthorectified aerial photographs

- provide high -resolution orthophotography for the U.S. every 5 to 10 years
- High -resolution orthorectified images combine the image characteristics of an aerial photograph with the geometric qualities of a map
- An orthoimage is a uniform -scale image where corrections have been made for feature displacement, such as building tilt and for -scale variations caused by terrain relief, center geometry, and camera tilt
- A digital ortho image may be created from several photographs, mosaic, to form the final image

US national map

- one of the cornerstones of the USGS National Geospatial Program, the National Map is a collaborative effort among the USGS and other federal, state, and local partners to improve and deliver topographic information for the nation
- provides a high-quality online map app for public use

National elevation dataset or NED

- is a primary elevation data product that has been produced and distributed by the USGS
- The USGS produces DEMs with 3, 10, and 30 meter horizontal sampling frequency and contour shape files as part of the National Elevation Dataset

National hydrography dataset

- represents the water drainage network of the United States with features such as rivers, streams, canals, lakes, ponds, coastline, dams, and stream gauges
- As of October 1st, 2023, the NHD was retired and replaced by the 3D Hydrography Program, or 3DHP

3D hydrography program

- replaces the National Hydrography Dataset, or NHD, the Watershed Boundary Dataset, and the NHD Plus High Resolution with a single product that includes lakes, streams, catchments, and drainage areas, as well as other hydrologic features

US archives and records administration

- provides old images of the U.S.

National aerial imagery program or NAIP

- provides photographs of the growing season for the US

- It's administered by the US Department of Agriculture
- Imagery resolution was one meter until 2017 and 0 .6 and 0 .3 meters starting in 2018
- Image tiles are available in 3 .75 minute, quarter quadrangle increments
- Images are natural color with red, green and blue bands or color near infrared with red, green, blue and near infrared bands and may contain as much as 10% cloud cover per tile.

National landcover database

- provides land cover data for the US from 2001 to 2021
- It is produced by the USGS
- Datasets include land cover, land cover change index and urban imperviousness

National agricultural statistical service or NASS

- studies and provides the market with detailed information about U .S. agriculture
- It annually produces a crop data layer

National wetlands inventory or NWI

- produces and distributes maps and other geospatial data on American wetland and deepwater habitats, as well as monitors changes in these habitats through time
- This information is available to the public through two primary datasets, the Wetlands Status and Trends Reports
- The National Wetlands Inventory is produced by the U .S. Fish and Wildlife Service

STATSGO2

- The Digital General Soil Map of the United States, or StatsGo2 database
- It is the Digital General Soil Map of the United States
- is a broad -based inventory of soils and nonsoil areas that occur in a repeatable pattern on the landscape and that can be cartographically shown at a scale of 1-250 ,000 in the continental US, Hawaii, Puerto Rico, and the Virgin Islands, and at a scale of 1-1,000,000 in Alaska
- The level of mapping is designed for broad planning and management uses covering state, regional, and multi -state areas
- It is administered by the USDA

SSURGO

- contains information about soil as collected by the National Cooperative Soil Survey over the course of a century
- The information can be displayed in tables or as maps and is available for most areas of the United States and the territories, commonwealths, and island nations served by the USDA
- The information was gathered by walking over the land and observing the soil
- Many soil samples were analyzed in laboratories
- The extent of the SIRGO datasets are county or multiple counties

GNATSGO

- The Gridded National Soil Survey Geographic Database, or GNATSCO, is a composite database that provides complete coverage of the best available soils information for all areas of the United States and island territories

- It was created by combining data from the Soil Survey Geographic Database, or SIRGO, and data from the State Soil Geographic Database, or STATSCO2, and Raster Soil Survey Database, or RSS. into a single seamless Esri file geodatabase

FEMA

- Provides floodplain maps for the U .S. The FEMA Flood Map Service Center is the official public source for flood hazard information produced in support of the National Flood Insurance Program, or NFIP

National climatic data center or NCDC

- maintains the world's largest climate data archive and provides climatological services and data to every sector of the United States economy and to users worldwide

United states census bureau

- maintains the TIGER system, or Topologically Integrated Geographic Encoding and Referencing System
- The TIGER system includes data for roads, hydrography, political boundaries, railroads, power lines, pipelines, parks, cemeteries, and addresses
- administers a major decennial national survey every 10 years, and the American Community Survey annually

American community survey or ACS

- provides statistics based on surveys from sampled individuals
- The Census conducts parts of the American Community Survey every year and other parts of it every five years

NOAA

- administers several initiatives to develop and distribute spatial datasets

National geodetic survey or NGS

- maintains foundational elements of latitude, longitude, elevation, and shoreline information
- hese elements impact a wide range of important activities. The NGS maintains geodetic datums, benchmarks, and the continuously operating reference system network

National ocean service or NOS

- is the country's leading authority on a wide variety of marine sciences, including hydrography, shoreline mapping, nautical charting, water level, tides, and currents measurement

Data Manipulation

501 Understanding of georeferencing, data format conversion, and data transformation

Georeferencing

- is associating a map without spatial information with spatial information
- can only be done with a projected coordinate system on a flat plane

- Unreferenced images in need of georeferencing are often scanned maps and aerial photography

Accurate georeferencing
- is accomplished by digitizing in the correct coordinate system, digitizing consistently, digitizing enough vertices, analyzing the root mean square error, and setting fuzzy tolerance and snapping appropriately

Coordinate transformation
- brings spatial data into an earth -based map coordinate system so that layers align
- is also known as registration
- commonly used to convert newly digitized data

Transformations
- are for georeferencing digitized data, not for converting between projections.

Affine transformation
- are first order polynomial transformations
- most commonly applied coordinate transformation
- An affine transformation can translate, rotate, scale, and skew the data
- they introduce less error than higher order polynomial functions
- are fit by minimizing the root mean square error
- A lower root mean square error indicates a more accurate transformation with affine transformations
- Analyzing the root mean square error is an iterative process
- The error is calculated for one control point at a time
- Points with the highest amount of error are corrected and the root mean square error is recalculated
- use six parameters and require at least three control points to perform, but four control points are needed to estimate the root mean square error

Similarity transformation
- is a subtype of a affine transformations
- can scale, rotate, and translate, they do not skew
- keep the original aspect ratio so shapes don't change
- can be done with just two control points

Conformal transformation
- is a first order polynomial transformation that requires equal scale changes in the x and y directions
- requires only two control points

Higher order polynomial transformation
- Higher order polynomial transformations are 2nd order, 3rd order, etc.
- Higher order polynomials allow more flexibility in warping the surface to fit the control points, but they can deform the non control point areas
- The root mean square error will typically be lower for higher order polynomial transformations, but they can introduce more error in the non -control point areas
- Root mean square error for higher order polynomial equations should be analyzed with caution
- Second order polynomial transformations require six control points

- Third order polynomial transformations require ten control points.

On-the-fly transformations

- Applying a transformation on the fly applies the transformation during the digitizing steps by overlaying a previously registered image

Rubbersheeting

- is a form of coordinate transformation that warps a vector dataset to match a known geographic space
- most commonly needed when a dataset has systematic positional error such as one digitized from a historic map of low accuracy
- normally uses polynomial equations

Transformation RMSE

- is useful for comparing the same types of transformations, but not different types of transformations
- For example, root -mean -square error can be used to compare an ethene to another ethene transformation, but shouldn't be used to compare an ethene to a third -order polynomial transformation

Projection transformation

- Transforms data from one map projection to another map projection
- Transforming data from one map projection to another is a three step process, first is to convert cartesian coordinates of source projection to geographic coordinates using inverse coordinates from current projection, second is performed datum transformation if the source and target projections use different datums, third is to project from the intermediary geographic coordinates to the target projection coordinates

Geographic transformation

- transforms coordinate locations between geographic coordinate systems such as NAD27, NAD83, and WGS84. A geographic transformation usually requires converting to a Cartesian XYZ coordinate system, and back to the geographic coordinate system
- often changes the data used

Three parameter method

- performs three linear shifts along each axis of a 3D Cartesian coordinate system

Seven parameter method

- performs three linear shifts along each axis 3 angular rotations, and a scale factor in a 3D Cartesian coordinate system

Molodensky method

- converts directly between two geographic coordinate systems without converting to a Cartesian coordinate model
- uses three or five parameters
- It performs three shifts, dx, dy, and dz, and adjusts the semi -major axis and the flattening of the spheroid

Helmert transformation

- is a datum transformation that uses 7 or 14 parameters

NADCON and HARN methods

- are grid -based methods used by the United States
- They divide the transformation area into cells

Datum transformations

- may be part of a projection transformation process, or may be used to transform between different geographic coordinate systems

VDATUM

- is a tool provided by the National Geodetic Survey to estimate conversions among datum transformations

3 Common Transformations

- Three common datum transformations are NAD27 to NAD83, NAD27 to WGS84, and NAD27 to WGS84

Control points

- are points with known coordinates for the digitized map and the target map projection
- are used to register unregistered data sets
- can also be used to estimate the error in transformations, usually by analyzing the root mean square error
- Sources for control points can be surveys, GPS, cardametric maps, or existing registered layers

Format conversion

- refers to converting data from one form to another, such as from vector to raster
- Raster to vector conversion is not difficult because you can just use the center point of each pixel as points for the features
- vector to raster conversion is difficult because pixels can distort lines

Data transformation

- can refer to moving data from one data structure to another data structure
- can involve processes such as data munging

Data munging

- is transforming data from a raw form to an organized form for analysis

Data integration

- is combining data from different sources and providing users with a unified view of them
- Data integration is often done to create a data lake or data warehouse

Data lake

- is a collection of raw, unorganized data

Data warehouse

- is a central repository of integrated, organized, and structured data from one or more disparate sources

502 Understanding of spatial data generalization operations and methods

Generalization

- generalization reduces detail in data
- Used to declutter maps, increase map draw speed, and hide sensitive information.
- is needed on small scale maps where there is not enough room to display all the features or details
- reduces data accuracy and causes information loss

Vector generalization

- techniques include, select, simplify, smooth, symbolize, merge, aggregate, typify, collapse, reclassify, exaggerate, displace, enhance, hierarchy, edge matching, topological simplification, and scale -dependent rendering

Select

- Select keeps some features and omits others.
- Select is also known as omit.

Simplify

- removes vertices in lines and boundaries.

Smooth

- makes the shapes look smoother, which may require adding vertices

Symbolization

- represents features with icons

Merge

- combines adjacent features
- also known as dissolve, agglomerate, and combine

Aggregate

- merges multiple features into a new composite feature
- often increases the dimension of features
- also known as combine, regionalize, and fuse

Typify

- replaces a large set of features with a smaller number of features
- For example, representing 10 points with just 3 or 4 points
- Also known as distribution refinement

Collapse

- reduces the dimension of a feature
- For example, representing a city with a point
- also known as symbolize and sequelitization

Reclassify

- reduces a large range of values into a small number of categories
- is also known as categorization

Exaggerate

- makes features larger than they are
- For example, representing roads with lines thicker than the roads really are on a small scale map
- Exaggerate is also known as emphasis

Displace

- moves the location of features that are too close together
- also known as conflict resolution

Enhance

- adds symbols or other details
- For example, adding a bridge symbol to a road to show a road overpass

Hierarchy

- organizes features into hierarchical levels

Edge matching

- makes boundaries between adjacent map sheets or tiles seamless

Topological simplification

- removes unnecessary topological details

Scale dependent rendering

- adjusts feature visibility based on the map scale

Raster generalization

- Raster datasets can be generalized by reducing the resolution
- Multiple resolutions can be stored in a raster pyramid

Resolution reduction

- decreases the spatial resolution of raster data
- results in a larger cell size, smaller file, faster display, a loss of accuracy, and information loss

Raster pyramids

- generalize a raster to improve display performance
- They are a down sampled version of the original raster dataset. and can contain many downsampled layers at different scales

503 Understanding of spatial file types and their applications and limitations

Shapefiles

- Vector file
- Industry standard for file based vector spatial data
- Consists of feature geometry, attribute data, and projection metadata
- Each shapefile can contain only one type of vector data type (point/line/poly)
- Shapefile extensions include shp which is feature geometry, shx stores index for fast retrieval, dbf stores attribute information, prj stores projection information, sbn and sbx are optional spatial index files for optimizing spatial queries, shp xml stores geospatial metadata in xml format
- Must have shp,dbf, and shx file to work

Geodatabases

- 3 types of geodatabase

File geodatabase

- Object model based spatial database containing schema and rules
- Hybrid and can contain vector, raster, and tabular data along with topologies, file attachments, and relationships
- Normally created by a single user and used by one or a few users
- Gdb and fgdb are file extensions

Personal geodatabase

- Datasets are stored within a microsoft access data file with is limited in size to 2 gigabytes
- Original database format and extension is mdb

Enterprise geodatabase
- Based on oracle, sql server, IBM db2, IDM informatix, or postgresql relational databases
- Provides additional capabilities of a relational database system which supports versioning and integration with other database systems
- For organizations with many users
- Functions for backup, recovery, replications, and sql support
- Uses arc sde as middleware between the relational database managements system and arcgis applications

GEOJSON
- Encodes geographic structures such as points, lines and polygons using javascript object notation and is widely used for web mapping applications
- File type is maintained by a group of developers
- .geojson or .json are file extensions

Topojson
- Extension of a geojson that encodes geospatial topology and provides smaller file sizes
- .topojson is file extension

Computer aided design
- Generated by specific design software such as autocad or microstation to represent 2d or 3d detailed real world objects
- Typically employed in design, engineering, architecture, surveying, and construction
- Many applications can import or export cad data formats
- Cad data doesn't have to be spatially referenced to an earth based spatial reference system like GIS
- .dpg, .dxf, and .dgn most popular file types

Drawing or dwg file
- Drawing file type is made with autocad
- Supports more complex information than dxf

Drawing exchange format file type or dxf file
- Made with autocad
- More widely compatible across software platforms than .dwg
- No topology but lots of details

Design or dgn file
- Made by microstation and supported by autocad

Digital line graph or dlg file
- Topologically structured vector feature maps developed by USGS
- Include nine different categories of features such as public land survey system, boundaries, transportation, hydrography, hypsography, non vegetative features, survey control markers, man made features, and vegetative surface cover
- Derived from USGS topographic maps
- Last made in 1990

Tiger
- Used by census bureau to describe land attributes such as roads, building, rivers, lakes, and census tracts
- developed to support the census

- Do not contain census demographic data but merely geospatial map data
- Demographic data stored in census can be combined with tiger files
- .ft# and rt# are file extensions

Scalable vector graphics
- Xml based vector image format for defining two dimensional graphics
- Supports interactivity and animation
- Open standard developed by world wide web consortium
- Can be scaled in size without loss of quality and SVG files can be indexed, scripted and compressed
- XML text files can be created and edited by text editors of graphic text editors and rendered by most web browsers
- .svg and .svgz are file extensions

ArcInfo coverage
- old file type that stores vector data
- Data is stored as a directory within which each feature class is stored as a set of files
- contain spatial data, attribute data, and they can have topology
- was replaced by the feature class
- file extensions are .adf, .rat, .sec, .PAT, .TXT, .NIT, .DAT, and .DIR

Arcinfo interchange file
- used to transfer vector data between machines
- uses a .EOO extension that increases to .EOO1, .EOO2, etc
- composed of several different files

Portable document format or PDF
- can store vector or raster image data
- standard format for publishing documents and maps

Enhanced metafile or EMF
- used by the graphic design industry for printing
- can store high quality vector and raster data
- created to replace the Windows Metafile, or WMF
- extension is .EMF

Geography markup language or GML
- is an XML -based language for expressing geographic features developed by the Open Geospatial Consortium, or OGC
- modeling language for geographic systems, as well as an open -interchange format for geographic transactions on the Internet
- can store vector and raster data
- uses a tag -based structure similar to HTML

Keyhole markup language or KML
- stores information about how data is visualized, whereas GML stores information about the locations and attributes of geographic data
- was developed for use in Google Earth and became a standard of the Open Geospatial Consortium, or OGC, in 2008
- uses a tag -based structure similar to HTML
- extension is .KML

Keyhole markup language zipped or KMZ

- is a compressed file format used for storing geographic data, including KML files as well as associated resources, such as photos, icons, and other files
- used for storing geographic data, such as photos, icons, and other files
- used to bundle and transmit vast amounts of geographic data in a single file
- Extension is .kmz

Postscript or PS

- is a page -definition language used in electronic publishing and desktop publishing
- a programming language that can be used for a variety of purposes, including defining vector graphic files
- extension is .ps

Well known text or WKT

- language for representing vector geometry objects defined by the OGC in their simple feature access
- extension is .WKT

Tagged image file format or TIFF

- is a lossless image file format used for storing raster graphics
- commonly used by photographers, graphic artists, and the publishing industry
- known for their high quality, flexibility, and near universal compatibility
- originally developed by the Aldis Corporation for scanners in the 1980s
- extensions are .TIFF and .TIF

Geotiff

- is a public domain metadata standard that allows geo -referencing information to be embedded within a TIFF file
- a sidecar file for the TIFF file type
- can contain information about map projections, coordinate systems, ellipsoids, datums, and more
- OGC published Geotiff standards in 2019
- extension is .TFW

Joint photographic experts group or JPEG

- file type is the most widely used method of lossy compression for digital images
- degree of compression can be adjusted, allowing a selectable trade -off between storage size and image quality
- typically achieves 10 to 1 compression with little perceptible loss in image quality
- extensions are jpg, jpeg, jpe, jif, jfif, and jfi

JPEG 2000

- is a newer version of JPEG
- extensions are jp2 and jpx

Image or IMG

- several file types
- One file type used in GIS is a planar bitmap graphics file using simple run length encoding, used by the Earth Resources Data Analysis System for aerial imagery
- More commonly, IMG file types are used for storing disk images of hard drives
- file extension is .IMG

ArcGIS grid

- is a proprietary format for raster datasets developed by Esri
- It has two formats, ArcInfo Grid and ArcInfo ASCII Grid

Network common data form or NetCDF

- set of software libraries and self -describing machine -independent data formats that support their creation, access, and sharing of array -oriented scientific data
- used for multidimensional scientific data such as climate models

Hierarchical data format or HDF

- is a set of file formats such as HDF4 and HDF5
- was originally developed at the U.S. National Center for Supercomputer Applications to store and organize large amounts of data
- extensions begin with .HDS

Graphic interchange format or GIF

- is a digital file format for images and short animations that was created in 1987 by CompuServe
- are raster files that can support up to 256 colors and 8 bits per pixel
- can be used to combine images or frames to create basic animations
- are popular for transmitting and storing graphic files because they use a lossless static compression format called LZW, which means no information is lost during compression
- This results in small file sizes that make GIFs easy to share and load quickly on websites and social media platforms
- graphic interchange format extension is .GIS

USGS digital elevation model or DEM

- is a geospatial file format developed by the United States Geological Survey for storing a raster -based digital elevation model
- It is an open standard and is used throughout the world
- has been superseded by the USGS's own SDTS format, but the format remains popular due to large numbers of legacy files, self -containment, relatively simple field structure, and broad, mature software support
- Extension is .dem

Tape archive file or TAR

- combines multiple files into a single archive for backup or distribution
- The Landsat imagery is initially stored in zip TAR files by the USGS
- TAR.gz is a zipped TAR file
- file extensions are .TAR and .TAR.gz

LIDAR aerial survey or LAS

- aerial survey file type is a widely used binary format designed to store 3D point cloud data collected by LIDAR surveying systems
- Each LAS file contains a collection of points with X, Y, and Z coordinates, attributes, intensity values, return numbers, and classification codes
- The LIDAR aerial survey file extension is .LAS

LAS Zipped or LAZ

- is a compressed version of the LAS format
- developed in 2007 as an open source solution

- reduces the file size of LAS files while retaining all the original data
- extension is .LAS

Enhanced compression wavelet or ECW

- is a proprietary, glossy wavelet compression image format used for aerial photography and satellite imagery owned by Intergraph
- Georeferencing information can be embedded in the file
- extension is .ECW

Multiresolution seamless image database or MRSID

- was developed and patented by LizardTek for encoding of georeferenced raster graphics
- It can be lossy or losslessly compressed from Geotiff, ACW, LAS, and other image file types
- encode multiple resolutions that work like pyramids but without creating extra files
- When you view the file, it only decodes the resolution level that you need
- extension is .SID

Raster auxiliary files

- store information that a raster cannot store, such as statistics, histograms, tables, pointers, etc and projection information in a sidecar file
- extensions are .aux and .aux.xml

World file

- is a six line plain text sidecar file used by geographic information systems to georeference raster map images
- file specification was introduced by Esri
- can be used with JIF, JPEG, JPEG 2000, PNG and TIFF files
- World files override geographic data stored in the header of accompanying files
- extensions are .gfw, .jgw, .j2w, .pgw, and .tsw

Raster catalog

- isn't a file type
- It is an Esri tool for storing a collection of raster data sets defined in a table in which the records define the individual raster data sets that are included in the catalog
- can be used to display adjacent or overlapping raster data sets without having to mosaic them together into one large file.
- Geodatabases support raster catalogs

Mosaic dataset

- A mosaic data set is an Esri raster catalog used to manage, display, analyze, serve, and share imagery and raster data
- Mosaics can create a continuous image surface across a large area
- Mosaic data sets can store multi -dimensional collections of time series layers
- Geodatabases support mosaic data sets

Geopackage

- an open universal format for storing geodata based on SQLite
- defined by the Open Geospatial Consortium or OGC
- can store vector and raster data
- It is an SQLite container

- Geopackage Encoding Standard governs rules and requirements of content stored in a geopackage container
- there are geopackage standards for storing vector features, pile matrix, raster maps, attributes, and extensions
- extension is .GPKG

American Standard Code for Information Interchange or ASCII

- standard way of encoding characters as 7 -bit strings of zeros and ones
- 3 common ASCII file types are DAT, CSV, and TXT

Band interleaved by pixel or BIP

- one of the three primary methods for encoding image data for multi -band raster images in the geospatial domain, such as images obtained from satellites
- BIP is not, in itself, an image format, but it is a method for encoding the actual pixel values of an image in a file
- Images stored in BIP format have the first pixel for all bands in sequential order, followed by the second pixel for all bands, followed by the third pixel for all bands, etc

Band interleaved by line or BIL

- one of the three primary methods for encoding image data for multi -band raster images in the geospatial domain, such as images obtained from satellites
- is not in itself an image format, but it is a scheme for storing the actual picture values of an image in a file, band by band for each line or row in the image
- For example, given a three band image, all three bands of data are written for row one, then all three bands of data are written for row two, and so on and so forth

Band sequential format or BSQ

- one of three primary methods for encoding image data for multi -band raster images in the geospatial domain, such as images obtained from satellites
- is not in itself an image format, but is a method for encoding the actual pixel values of an image in a file
- is a very simple format, where each line of the data is followed immediately by the next line in the same spectral band
- optimal for spatial or XY access of any part of a single spectral band

Header georeferencing

- Some files store georeferencing information in the header, such as ERDAS, B -S -Q, B -I -L, B -I -P, Geotiff, and Grid files

Freeman codes

- are a way of encoding lines for vector files
- Lines are coded with a chain of numbers, such as 0 through 7, each number represents a different direction

Run length encoding

- is a lossless compression method where sequences that display redundant data are stored as a single data value
- is effective for compressing files with large homogeneous areas

LZ77 and LZ78

- are lossless pyramid data compression algorithms created by Abraham Lapel and Jacob Ziv

- are used for the basis of many other LZ compression algorithms
- are dictionary encoders that use a sliding window compression to replace repeated occurrences of data with references to a single copy of that data

504 Understanding of data integration

Data integration
- Combines data from multiple sources into one unified view

Concatenation
- Integration of two or more different data sources such that all the contents of each are accessible in the product

Conflations
- Replaces two or more version of the same information with a single version that reflects the pooling or weighted averaging of the sources

Extract, transform, load (ETL)
- Standard methodology for integrating data across systems
- Originated with the emergence of relational databases that stored data in tables for analysis
- ETL tools were developed to automatically convert transactional data into relational data with interconnected tables
- Basic methodology is three step process, first extract data from source, second transform data into a format used by the target system, and third load data to the final storage point and format

Data pipeline
- Method where raw data is ingested form data sources, transformed and then stored into a data lake or data warehouse for analysis
- Basically a diagram for an ETL workflow

Data warehouse
- Central repository of integrated data from one or more disparate sources
- Purpose is to support various data related activities including analysis and report creation to help decision makers draw insight and make informed decision
- Often created from an ETL pipeline

Analytical Methods

601 Understanding of data selection queries and views

Attribute Selection
- is selecting features based on filters of attributes
- types of attribute selection include new selection, add to selection, remove from selection, subset selection, switch selection, and clear selection

Spatial selection

- include intersect within a distance, contains, which is when features contain an input polygon

Completely contains

- which is when features must be completely in an input polygon, the input polygon is selected.

Contains Clementini.

- Features must be completely in the input polygon, but if it's on the boundary, it will not be selected

Within

- Features will be selected if inside a selecting polygon

Completely within

- features will be selected if completely within selecting polygon, with no overlap

Within Clementini

- is when features will be selected and cannot be entirely on the boundary of the features

Are Identical To

- Features are identical to the input layer

Boundary touches

- Features will be selected if they have have a boundary that touches selecting features
- They must be completely inside or outside the polygon.

Share a line segment with

- is when features are selected if they share a line segment

Crossed by the outline of

- is when input features will be selected if they are crossed by the outline of a selecting feature

Have their center in

- Features will be selected if their center falls within a selecting feature

Contained by

- is the same as within

Location selection

- Selecting features within a certain distance of a location

Spatial data selection functions

- **Join**
    - combining two attribute tables into one using a common key between tables
- **Merge**
    - combines multiple input data sets of the same data type into a single new output
- **Append**
    - combines data sets of the same data type into an existing data set
- **Union**
    - combines input features with another feature data set
- **Clip**
    - extracts input features that overlay the clip features and keeps inputs attributes
- **Intersect**
    - extracts features which overlap all in all layers to a new feature class and joins the attribute tables

602 Understanding of techniques and implications of data classification

Classification

- is grouping objects with similar symbols or the process of sorting or arranging entities into groups or categories
- Up to seven classes are recommended, but sticking to five classes is better
- Classes should be exhaustive, which means they describe all possible values, and should not overlap, means no value can fall into two classes
- There are different kinds of classifications, but all will generally involve a classification schema or key, which is a set of criteria, usually based on the attributes of the individuals, for deciding which individuals go into each class.

Types of Data classification

- include equal range, quantiles, standard deviation, natural breaks, and symbology

Data Classification Techniques

- are equal range, which means equal distance between class breaks, quantiles, which means an equal number of observations in each class. Standard deviation makes the class breaks based on distance of standard deviation from the mean. Natural breaks is when the class breaks conform to gaps in data distribution. Symbology is when one layer can be symbolized by an attribute.
- The implications of using different classification styles is that each will highlight different aspects of the trends of the dataset
- Choosing the wrong classification methodology could cause the data to hide important information or misinform the reader

Data classification

- combines raw data into predefined classes or bins
- These classes may be represented in a map by some unique symbols, or, in the case of choropleth maps, by a unique color or hue

Choropleth Map

- are thematic maps shaded with graduated colors to represent some statistical variable of interest

Equal Interval or Equal Step

- divides the range of attribute values into equally sized classes

Quantile Breaks

- produces equal numbers of observations into each class
- This method is best for data that is evenly distributed across its range

Natural Breaks or Jenks

- utilizes an algorithm to group values in classes that are separated by distinct breakpoints
- is best used with data that is unevenly distributed, but not skewed toward either end of the distribution

Standard Deviation

- forms each class by adding and subtracting the standard deviation from the mean of the dataset
- This method is best suited to be used with data that conforms to a normal distribution

603 Understanding of analytical operations and methods

Overlay Analysis
- based on Boolean logic, includes first define the problem, second break the problem into sub -models, third determine significant layers or create layers as necessary, fourth reclassify or transform data within a layer

Spatial Overlay
- is the process of superimposing layers of geographic data that cover the same area to study the relationship between them

Overlay
- is the process of taking two or more maps or layers and superimposing them for showing relationships between the features

Vector Overlay Tools
- **Identity**
  - is taking input features and splitting them by overlay features
- **Intersect**
  - is selecting only features common to all input layers.
- **Symmetric Difference**
  - is selecting features in common to either the input layer or the overlay layer but not both
- **Union**
  - is selecting all input features
- **Update**
  - is inputting feature geometry replaced by an update layer

Raster Overlay Tools
- **Zonal Statistics**
  - summarizes values in a raster layer by zones or categories in another layer
  - For example, calculate the mean elevation for each vegetation category
- **Combine**
  - assigns a value to each cell and output layer based on unique combinations of values from several input layers
- **Single output map algebra overlay**
  - lets you combine multiple raster layers using an expression you enter.
  - For example, you can add several ranked layers to create an overall ranking
- **Weighted Overlay**
  - automates the raster overlay process and lets you assign weights to each layer before adding.
  - you can also specify equal influence to create an unweighted overlay
- **Weighted Sum**
  - overlays several rasters multiplying each by their given weight and summing them together

Understanding Overlay Analysis
- is a group of methodologies applied in optimal site selection or suitability modeling

- It is a technique for applying a common scale of values to diverse and dissimilar inputs to create an integrated analysis
- Suitability models identify the best or most preferred locations for a specific phenomenon
- often requires the analysis of many different factors
- For instance, choosing the site for a new housing development means assessing such things as land cost, proximity to existing services, slope, and flood frequency

General Steps for Overlay Analysis
- Define the problem
- Break the problem into submodels
- Determine significant layers
- Reclassify or transform the data within a layer
- Weight the input layers
- Add or combine the layers
- Analyze

Fuzzy Membership
- reclassifies or transforms the input data to a 0 to 1 scale based on the possibility of being a member of a specified set
- 0 is assigned to those locations that are definitely not a member of the specified set
- is assigned to those values that are definitely a member of the specified set
- The larger the number, the greater the possibility.

Fuzzy Overlay
- allows the analysis of the possibility of a phenomenon belonging to multiple sets in a multi -criteria overlay analysis
- Not only does fuzzy overlay determine what sets the phenomenon is possibly a member of, it also analyzes the relationships between the membership of the multiple sets

Overlay Type Lists
- Method combines the data based on set theory analysis. Each method allows the exploration of the membership of each cell belonging to various input criteria
- **Fuzzy And**
  - will return the minimum value of the sets the cell location belongs to
- **Fuzzy Or**
  - will return the maximum value of the sets the cell location belongs to
- **Fuzzy Product**
  - will, for each cell, multiply each of the fuzzy values for all the input criteria
- **Fuzzy Sum**
  - will add the fuzzy values of each set the cell location belongs to
- **Fuzzy Gamma**
  - is an algebraic product of fuzzy product and fuzzy sum, which are both raised to the power of gamma

604 Knowledge of map algebra

Map Algebra

- are the various functions performed on cells based on neighboring cells for raster data sets
- is an algebra for manipulating geographic data, primarily fields
- is raster math
- It is a set of primitive operations in a geographic information system, which allows one or more raster types of similar dimensions to produce a new raster layer using mathematical or other operations such as addition, subtraction, etc
- most common type of map algebra is a cell by cell function

Local Operations
- combine raster and raster. that overlay each other
- They add, subtract, etc., the cells that are in the same location
- use map algebra on a cell by cell basis

Global Operations
- apply a formula to all cells
- apply a bulk change to all cells in a raster
- They add, subtract, multiply, etc., all cells based on one value
- They can find the distance from one cell to all cells

Focal Operations
- calculates a value based on all neighboring cells
- can find the average of all the cells around a certain cell

Zonal Operations
- compute a value based on cells in a certain area
- apply a math function to a group of cells within a specified zone

Mathematical Map Algebra Functions
- Arithmetic operations include addition, subtraction, multiplication, and division
- Statistical operations include minimum, maximum, average, and median
- Relational operations include greater than, smaller than, and equal to
- Trigonometric operations include sine, cosine, tangent, and arcsine
- Exponential and logarithmic operations include exponent and logarithm

605 Knowledge of descriptive and spatial statistics

Descriptive Statistics
- is the discipline of comparatively describing the main features of a collection of information
- summarizes a sample to learn about the population
- is a summary statistic that quantitatively describes or summarizes features from a collection of information

Summary Statistics
- is used to summarize a set of observations

Coefficient of Determination
- uses r -squared, a number that indicates how well data fit in a statistical model
- r -squared summarizes the populations fit to a line or a curve

- 1 indicates the line fits perfectly with the data
- 0 indicates the line does not fit at all and that the data is random

Non -parametric statistics

- is the type of statistics that is not restricted by assumptions concerning the nature of the population from which a sample is drawn

Parametric Staticts

- a problem is restricted by assumptions concerning the specific distribution of the population

Measures of central tendency

- include the mean, median, and mode

Measure of variability

- include the standard deviation or variance
- the minimum and maximum values of the variables, kurtosis, and skewness

Central tendency

- is a central or typical value for a probability distribution

Dispersion

- Dispersion is the extent to which a distribution is stretched or squeezed

Standard Deviation

- is a measure of the amount of variation or dispersion of a set of values
- A low standard deviation indicates that the values tend to be close to the mean of the set, while a high standard deviation indicates that the values are spread out over a wider range

Variance

- is the squared deviation from the mean of a random variable
- meaning it is a measure of how far a set of numbers is spread out from the mean

Kurtosis

- is a measure of the tailedness of the probability distribution of a real valued random variable

Skewness

- is a measure of the asymmetry of the probability distribution of a real valued random variable about its mean

Univariate analysis

- involves describing the distribution of a single variable

Bivariate and multivariate analysis

- When a sample consists of more than one variable, descriptive statistics may be used to describe the relationship between pairs of variables. Descriptive statistics include cross tabulations and contingency tables, graphical representation via scatter plots, quantitative measures of dependence, and descriptions of conditional distributions.
- The main reason for differentiating univariate and bivariate analysis is that bivariate analysis is not only a simple descriptive analysis, but also it describes the relationship between two different variables.

Contingency Table

- also known as a cross tabulation or cross tab, is a type of table in a matrix format that displays the multivariate frequency distribution of the variables

Scatter Plot

- is a type of plot or mathematical diagram using Cartesian coordinates to display values for typically two variables for a set of data

Correlation or Dependence

- is any statistical relationship whether casual or not, between two random variables or bivariate data
- A correlation may indicate any type of association. It usually refers to the degree with which a pair of variables are linearly related

Conditional probability distribution

- Given two jointly distributed random variables, the conditional probability distribution of x given y is the probability distribution of x when y is known to be a particular value

Slope of Regression

- reflects the relationship between variables

Standardized Slope

- indicates this change in standardized or z -score units

Pearson Correlation Coefficient or Pearson's R

- a correlation coefficient that measures linear correlation between two sets of data
- It is the ratio between the covariance of two variables and the product of their standard deviations
- Thus it is essentially a normalized measurement of covariance such that the result will always be between negative one and one

Covariance

- is a measure of the joint variability of two random variables
- if the greater values of one variable mainly correspond with greater values of the other variable, and the same holds for the lesser values, the covariance is positive
- in the opposite case, when the greater values of one variable mainly correspond to the lesser values of the other, the covariance is negative

Prediction

- is a statement about a future event or data

Z- Score

- or standard score, is the number of standard deviations by which the value of a raw score is above or below the mean value of what is being observed or measured
- Raw scores above the mean have positive standard scores while those below the mean have negative standard scores

Coefficient of determination

- The coefficient of determination denoted R -squared and pronounced R -squared is the proportion of the variation of the dependent variable that is predictable from the independent variables

Simple Linear Regression

- is a linear regression model with a single explanatory variable
- That is, it concerns two -dimensional sample points with one independent variable and one dependent variable, and finds a linear function, a non -vertical straight line, that as accurately as possible predicts the dependent variable

Database Design and Management

701 Understanding of relationships among database objects

Schema
- is the structure or design of the database or database object
- can include tables, views, indexes, stored procedures, and triggers
- defines the tables, the fields in each table, and the relationships between fields
- will include information on which fields have domains and what those domains are

Data Dictionary
- is a catalog or table containing information about the data set stored in a database

Domain
- s the range of values for a particular metadata element.

Attribute Domain
- enforces data integrity and identifies which values are allowed in a field in a feature class

Coded Value Domain
- is an attribute domain that identifies a set of permissible values for an attribute in a geodatabase
- It has a code and its equivalent value

Range Domain
- A range domain is an attribute domain
- is a type of attribute domain that identifies the range of permissible values for a numeric attribute

Spatial Domain
- is the allowable range for x and y coordinates and for m or z values

Tables
- are a collection of related data held in a structured format within a database
- contain fields and rows

Views
- are a result of a set of queries on the data
- Users can query the data to produce a view
- A virtual table can be computed dynamically from data when the view is accessed

Sequences
- are ordered collections of objects in which repetitions are allowed
- The number of elements is the length of the sequence

Synonyms
- are an alias or alternate name for a table, view, sequence, or other object

Indexes
- are a data structure that improves the speed of data retrieval operations in a database table
- cause more storage space and additional writes
- They quickly locate data in the database

Clusters

- can be either multiple servers share one storage, This is typically used to handle user loading balancing

Databases distributed to different servers using replication

- **T**his is typically used if you have multiple users utilizing the same data in different physical locations
- There is a master database that the replica databases sync between

Database Links

- are data stored in different databases but accessible by the database currently being accessed

Snapshot

- is the state of a system at a particular point in time
- can be used as a backup

Procedure

- is a subroutine available to applications that access a relational database system
- are used for data validation and access control mechanism

Trigger

- is procedural code automatically executed in response to certain events on a particular table or view in a database

Functions

- are sequences of program instructions that perform a specific task
- are subroutines

Package

- is built from a one of the available package management systems

Non-Schema Objects

- include users, roles, contexts, and directory objects

Database Schema

- is the structure of a database described in a formal language supported typically by a relational database management system
- term schema refers to the organization of data as a blueprint of how the database is constructed
- The formal definition of a database schema is a set of formulas or sentences called integrity constraints imposed on a database
- These integrity constraints ensure compatibility between parts of the schema. All constraints are expressible in the same language

Relational Database

- s a database based on the relational model of data as proposed by E .F. Codd in 1970
- E.F. Codd termed the relational database a relational model of data for large shared data banks
- A relational model organizes data into one or more tables or relations of columns and rows with a unique key identifying each row
- Rows are also called records or tuples
- Columns are called attributes
- Rows in a table can be linked to rows in other tables by adding a column for the unique key of the linked row

- Relations that store data are called base relations and implementations are called tables

Constraints
- are often used to make it possible to further restrict the domain of an attribute
- Constraints provide one method of implementing business rules in the database and support subsequent data use within the application layer

Primary Key
- Every relation or table has a primary key
- uniquely specifies a tuple within a table

Foreign Key
- refers to a field in a relational table that matches the primary key column of another table
- It relates the two keys

Stored Procedures
- is an executable code that is associated with and generally stored in a database
- usually collect and customize common operations like inserting a tuple into a relation, gathering statistical information about usage patterns, or encapsulating complex business logic and calculations
- Frequently they are used as an Application Programming Interface, or API, for security or simplicity

Index
- An index is one way of providing quicker access to data
- Indices can be created on any combination of attributes in a relation
- Queries that filter using those attributes can find matching tuples directly using the index without having to check each tuple in turn

Relational Operations or Relational Algebra
- Queries made against the relational database and the derived revlars in the database are expressed in a relational calculus or relational algebra

Union Operator
- combines the tuples of two relations and removes all duplicate tuples from the result

Relational Union Operator
- is equivalent to the SQL union operator

Intersection Operator
- produces the set of tuples that two relations share in common
- intersection is implemented in SQL in the form of the intersect operator

Set Difference Operator
- acts on two relations and produces the set of tuples from the first relation that do not exist in the second relation
- Difference is implemented in SQL in the form of the accept or minus operator

Cartesian Product
- The Cartesian product of two relations is a join that is not restricted by any criteria, resulting in every tuple of the first relation being matched with every tuple of the second relation
- is implemented in SQL as the cross-join operator

Relational Division Operator

- is a slightly more complex operation and essentially involves using the tuples of one relation, the dividend, to partition a second relation, the divisor

702 Understanding of database design

Database Design
- is the process of producing a detailed data model of a database
- The design process includes conceptual schema, logical data model, and physical database design

Conceptual Schema
- you determine where relationships and dependency is within the data

Logical Data Model
- arranges data in a logical structure that can be mapped into the storage objects supported by the database management system

Physical Database Design
- includes the physical configuration of the database on the storage media, the detailed specification of data elements, data types, indexing options, and other parameters residing in the database management system data directory

Modules, Hardware, and Software
- he three stages of database design are conceptual schema, logical data model, and physical database design

Field Types
- Proper field type will secure data and make databases more efficient

Short Integer
- can store numbers between negative thirty two thousa

Long Integer
- can store integers between negative two billion and positive two billion

Float
- is a single precision floating point numbers

Double
- is a double precision floating point number

Text
- could be a coded value

Date
- are a calendar date and sometimes a time

Blobs
- are data stored as long as a long sequence of binary numbers
- ArcGIS stores annotation and dimensions as blobs

Object Identifiers
- are unique IDs and FIDs

Global Identifiers
- are global ID and GUID
- global identifiers are data types stored in registry style strings consisting of 36 characters enclosed in curly brackets

Raster Field Type

- can be stored within a database

Geometry Field Type

- can include point line, polygon, multi -point, and multi -patch data types

Overview of Geodatabase Design

- Is based on a common set of fundamental GIS design steps
- involves organizing geographic information into a series of data themes or layers that can be integrated using geographic location
- begins with deciding what geographic representations will be used for each data set

703 Knowledge of database management and administration

Basic Tasks

- backup and recover databases, test backup and recovery plan, ensure backups are done on schedule

Database Security

- prevent hackers, security models, tasks authentication and authorization
- Auditing, make sure the right people have the right access

Storage and capacity planning

- disk storage is needed and monitored by GIS and watch growth trends

Performance monitoring and tuning

- Identify bottlenecks
- Tuning means using indexing, queries on the speed of return, write monitoring tools, capacity of server hardware, troubleshooting

High availability and ETL functions

- Data extraction, transformation, and loading

Archiving

- Captures, manages, and analyzes data changes

Retrieval

- is extracting data from a backup due to loss or data corruption

The geodatabase administrator in SQL Server

- the geodatabase administrator can be either a user named SDE or a login that is mapped to the DBO user in the database that contains the geodatabase
- is responsible for the administration of the geodatabase system tables, triggers, views, and procedures. The default geodatabase version

GIS Data Administration

- Data management today includes multiple publication formats to improve display performance and capture chains over time

Useful Data

- is what you use to create business information products and enable informed business decisions

Useless Data

- is what you do not use

GIS Feature Data Production Database

- A production data source is an enterprise geodatabase used to organize and manage your geospatial feature data resources

GIS Feature Data Publication Database

- A publication geodatabase is an enterprise or file geodatabase used to optimize distribution of finalized geospatial data resources.

GIS Feature Data Map Cache

- is a collection of preprocessed tile map images stored at multiple scales for rapid dissemination

ArcGIS Data Stores

- an Esri managed data repository that stores spatial content that has been shared to the portal

GIS feature data archiving

- Geodatabase archiving it includes functionality to record and access changes made to all or a subset of data in a geodatabase

Enterprise Database

- provides a way to manage long transaction edit sessions with a single database instance
- Enterprise supports long transactions using versions which are different views of the database.
- A geodatabase can support thousands of concurrent versions of the data within a single database instance

Versioning

- allows multiple users to edit the same data in an enterprise geodatabase without applying locks or duplicating data
- Users access an enterprise geodatabase through a version. When you connect to a multi-user geodatabase you specify the version to what you will connect

Traditional Versioning

- provides the flexibility to work within versions for long transactions when accessed directly from the enterprise geodatabase
- a simplified editing experience when using feature services to accommodate shorter transactions

Branch Versioning

- facilitates the WebGIS model by allowing multi -user editing scenarios and long transactions via feature services

704 Knowledge of data security

Data Owner

- A data owner is a user who creates tables, feature classes, and who owns those datasets

User Access

- Database must verify the user accounts that connect to it.
- An administrator has full control of the database, can read, create, and update, delete features, and can create and delete feature classes and tables
- An editor can read, update, create, and delete features

- A reader can only view data
- A creator can create additional feature classes, tables, as well as read, update, create, and delete features

Authentication

- checks the list of users to make sure a user is allowed to make a connection
- can take place on the operating system or on the database

Groups

- you can grant groups of users based on their common functions
- Public role, the right granted to anyone connected to a database

Database Administrator

- Data Integrity
- tasks include installation, configuration, upgrade, migration, backup and recovery, database security, storage and capacity planning, performance monitoring and tuning, and troubleshooting

Application Development

801 Knowledge of data transfer protocols

Transfer Constructs

- are based on data models.
- 1. Logical constructs solely pertaining to this standard.
- 2. Constructs relating to the implementation method.
- 3. Constructs solely pertaining to the transfer media. 4. A file -based transfer is when data is in a structured file format

Application Programming Interface, or API.

- is data accessed and exchanged as needed between software systems.

Web Services

- data is accessed and exchanged over networks and the internet between software components using HTTP and other web -based protocols

Transfer of data between a client server and an end user

- SSL stands for Secure Sockets Layer.
- TLS stands for Transport Network Transport Protocol
- SSI and TSL are used to encrypt data

Transmission control protocol

- is the header package for the data at the transport layer

Internet Protocol

- is when the header is added at the Internet layer

Media access control or MAC address

- is added to the physical network layer, transferred from the host to the receiver

Network file services or NFS

- common internet file system

HTTP Protocol

- are hypertext transfer protocols
- They are the standard web transmission protocol and a way for delivering map images or map data to web browsers

Protocols for packet switched networks

- transmits data that is divided into units called packets
- A packet comprises a header, which describes the packet and a payload, the data
- The internet is a packet switched network

Transport Layer Protocols

- Transmission control protocol or TCP
    - indicates which protocols the transfer protocol uses at the transport layer
- User datagram protocol or UDP

Encryption scheme

- applies to transmitted data. only and not to the authentication system

File transfer protocol or FTP

- is a standard communication protocol used for the transfer of computer files from a server to a client on a computer network
- is used to transfer files from one host to another regardless of the host's physical locations
- IT IS NOT SECURE
- is built on a client server model architecture using separate control and data connections between the client and the server

Network Communications

- provide the required connectivity for distributed GIS operations

Types of Networks

- Local Area Network, or LAN
    - provide high bandwidth communications for short distances
- Wide area networks or WAN
    - provide an infrastructure enabling communication between remote locations

Bandwidth

- The network capacity is identified as bandwidth

Network transport protocols

- The communication packet is constructed at different layers during the transmission process. These include the application layer, the transport layer, the internet layer, and the network access layer

ArcSDE Database Access Protocols

- includes client and server communication components
- The database management system includes intelligence to support query processing
- Data is decompressed when processed by an ArcGIS client

ArcSDE API, Application Server Connect Architecture

- Initially, SDE was developed to extend database management support for spatial data types
- A proprietary ArcSDE API was used by ArcGIS clients to connect to SDE installed on the database management system server
- The ArcSDE middleware functions are supported on the client platform

Database API, ArcGIS direct connect architecture

- ArcGIS client direct connect option connects with a database management system client application program interface executed on the client desktop
- The SDE translation functions are supported on the client platform connecting to the database management system client and the database management system network client supports data transmission to the server
- Query processing remains on the database management system server
- ArcGIS direct connect supports feature access. to a variety of supported database environments

Hypertext Transfer Protocol, HTTP

- is a standard web transmission protocol
- It is a transaction -based environment
- Service selection and display are controlled by the browser client
- Data is compressed for transmission

Network latency

- is the round trip travel time for a single communication packet
- Many applications require several sequential round trips to the server and back to complete the display transmission. Each round trip is a chat, and most applications require several sequential round trips to the server to complete a display transaction
- The chattier the application, the more latency will impact display performance
- Latency does not exist in the local network

ArcGIS Workflow Server Output Formats

- PEG is recommended for imagery output
- PNG8 is a lightweight portable network graphics lossless data compression supporting up to 256 colors and transparency
- PNG24 is a medium -weight portable network graphics lossless data compression supporting up to 16 million colors and transparency
- PNG32 is the highest quality portable network graphics lossless data compression. It provides the best support for graphics with gradients, varying colors, rounded edges, and transparency.
- PDF is Adobe's portable document format.

Independent computing architecture or ICA

- is the independent computing architecture protocol developed by Citrix for communications between remote desktop host and Citrix remote client terminal displays

Enterprise Geodatabase Formats

- include EGD, which is imagery stored in an enterprise geodatabase
- TIFF uncompressed is an uncompressed tagged image file format
- TIFF LZW compress, LZW is a lossless compression
- TIFF jpg compression is a lossy jpeg compression of a tiff file
- JPG2000 is an upgraded jpeg or joint photographic experts group compression algorithm published in 2000
- MRSID is a lizard -tech multi -resolution seamless image database format that uses a mix of techniques supporting lossless and lossy compressed datasets for use by the GIS community

- ECW is the Enhanced Compression Wavelet format optimized for aerial and satellite imagery
- IMG_ERDAS is an ERDAS image image file format

Network bandwidth suitability

- is an important part of the system design process
- Expanding GIS technology patterns, particularly over wide area and mobile cellular networks, have placed heavy demands on the network infrastructure
- Too often, we see clients deploy solutions for client locations with insufficient network infrastructure. These mistakes can be avoided by completing a network suitability analysis as part of your system architecture design

802 Knowledge of coding, scripting, and modeling basics

Scripting

- is interpreted by the computer rather than compiled
- is used to manipulate, customize, and automate existing software

Object-oriented programming, or OOP

- is a programming paradigm based on the concept of objects, which are data structures that contain data in the form of fields, or attributes, and code in the form of procedures, or methods

Extensibility

- is a system design principle where the implementation takes future growth into consideration
- includes the level of effort to extend the system and implement the extension

Query expressions

- select a subset of features or records

Scripting basics

- includes expression, variables, iterations, and condition statements
- **Expression**
  - is the most basic programming instruction
  - It can contain values and operators that can reduce to a single value
- **Variables**
  - are values that can change depending on the program or information passed to the program.
- **Iterations**
  - are for loops, do while loops, and do until loops
- **Conditions**
  - Condition statements include if, else, and else if
  - If is a conditional expression that is either true or false.
  - Else is combined with an if statement and if that statement is false defaults to the else condition
  - Else if is a check if a different condition from the first if is true

Significant object -oriented languages

- include Ada, ActionScript, C++, CommonLISP, and many more

Objects and classes

- Languages that support object -oriented programming typically use inheritance for code, reuse and extensibility in the form of either classes or prototypes
- Classes are the definitions for the data format and available procedures for a given type of class or object
- Classes may also contain data and procedures known as class methods
- Objects are instances of classes
- Each object is said to be an instance of a particular class

Methods

- Procedures in object -oriented programming are known as methods

Variables

- are known as fields, members, attributes or properties

Class variables

- belong to the class as a whole
- There is only one copy of each variable shared across all instances of the class

Instance variables

- or attributes, are data that belongs to individual objects.

Member variables

- refers to both the class and instance variables that are defined by a particular class

Class Methods

- belong to the class as a whole and have access to only class variables and inputs from the procedure call

Instance Methods

- belong to individual objects and have access to instance variables for the specific object they are called on

Extensibility

- is a software engineering and systems design principle that provides for future growth
- is a measure of the ability to extend a system and the level of effort required to implement the extension
- Extensions can be through the addition of new functionality or through modification of existing functionality
- An extensible system is one whose internal structure and data flow are minimally or not affected by new or modified functionality

Extensible design

- is to accept that not everything can be designed in advance

Python programming basics

- Expression is the most basic kind of programming instruction in the language
- Expressions consist of values, such as 2, and operators, such as plus, and they can always evaluate, that is, reduce down to a single value.
- The order of operations, also called precedents
- You can use parentheses to override the usual precedents if you need to
- A data type is a category for values, and every value belongs to exactly one data type

Python Data Types

- The integer, or int, data type indicates values that are whole numbers

- Numbers with a decimal point are called floating point numbers
- Python programs can also have text values called strings
- Values can be stored in variables. A variable is like a box in the computer's memory where you can store a single value

Assignment statements

- You can store values and variables with an assignment statement
- An assignment statement consists of a variable name, an equal sign, also called the assignment operator and the value to be stored

803 Awareness of basic application development methods

Types of application development

- include Agile, DevOps, Waterfall, Rapid Application Development, and Spiral Development

Agile Application Development

- is developing software in many increments of the new functionality
- Scrum, Crystal, Extreme Programming, and Feature Driven Development are types of Agile

DevOps

- is used for organizational changes that enhances collaboration between departments responsible for different segments of the development lifecycle
- DevOps methods benefit by significantly reducing time to market and improving customer satisfaction, product quality, and employee productivity and efficiency

Waterfall

- is an application development method with a rigid linear model that contains sequential phases. Requirements, design, implementation, verification, maintenance.
- Each phase must be complete before moving to the next phase
- is often slow and costly due to its rigid structure and tight controls

Rapid Application Development, or RAD

- is a condensed development process that produces high quality system with low investment costs
- contains four phases, requirements planning, user design, construction, and cutover
- user design and construction phases repeat until the user confirms the product meets all requirements
- is the most effective for projects with a well -defined business objective and clearly defined user group, but which are not computationally complex
- especially useful for small to medium projects that are time sensitive
- requires a stable team composition with highly skilled developers and users who are deeply knowledgeable about the application area

Spiral development

- is a combination of waterfall and rapid application development
- It combines the advantages of top -down and bottom -up concepts

Software development process

- is a process of planning and managing software development

- It typically involves dividing software development work into smaller, parallel, or small projects or sequential steps or sub -process processes to improve design and or product management
- It is also known as a software development life cycle
- A life cycle model is sometimes considered a more general term for a category of methodologies and a software development process is a more specific term to refer to a specific process chosen by a specific organization

Systems Design and Management

901 Knowledge of systems architecture and design, including various GIS softwares, platforms, and environments

Architecture design
- includes the requirements phase, the design phase, the construction phase, implementation phase, and capacity planning tool, or CPT
- Requirements phase includes a user needs assessment and workflow loads analysis, including a baseline and peak traffic measurement
- The design phase includes infrastructure requirements, network communication capacity, hardware and software procurement, software development, and data acquisition must be identified.
- The construction phase includes system procurement, data acquisition, and design, authorization for application design and development, and prototype testing
- The implementation phase includes initial deployment and operational testing, final system delivery, user training, and system maintenance operations.
- The capacity planning tool was developed as a framework to promote successful GIS system design and implementation

Enterprise Technologies
- include database servers, storage area networks, Windows terminal servers, web servers, map servers, and desktop clients. Platforms
- The enterprise GIS environment is a broad spectrum of technology integration of enterprise technologies connected by local area networks, wide area networks, and internet communications

Platforms
- Different platforms for GIS include a desktop which is an individual user on a computer making maps doing data analysis and data creation, a server

Server
- A server brings GIS into the hands of everyone in the organization allows access to web GIS, control of GIS data on your own infrastructure, control over how GIS platform is deployed, maintained, secured, and used

- Being hosted in the cloud gives GIS the ability to discover, use, make and share maps with any device anywhere, anytime, gives access to other users, the maps and data that you produce, and connects more people outside of the organization to share the latest maps, data, and ideas

Enterprise GIS

- is integrated throughout the entire organization so that a large number of users can manage, share, and use spatial data and related information to address a variety of needs including data creation, modification, visualization, analysis, and dissemination
- can utilize both hosted in the cloud and server but if the organizational data is not stored in the cloud and only data is accessed from the cloud. than it is not quite an enterprise system

Software

- Software runs on a variety of hardware types
- GIS software can run on centralized servers or desktop computers
- The software may rely on the database management type, the database management system type, or the operating system type.

System infrastructure

- can include hardware, software, and a communication network
- It includes required information about products and spatial and non -spatial data resources
- Essential spatial analysis, display and reporting functions, data management resources, and the anticipated number of end users within the department

Virtualization

- is the creation of a virtual machine that acts like a real computer with an operating system

Types of GIS software

- GRASS, which was developed by the U .S. Army Corps of Engineers, its open source, ESRI, which produces ArcMap and ArcGIS Pro, QGIS, MapInfo, and Small World

System Design Strategies

- The components of system design include the system design process, GIS software technology, software performance, server software performance, GIS data administration, network communications, platform performance, information security, GIS product architecture, performance management, system implementation, the capacity planning tool, etc

System Design Process

- The system design methodology includes identifying business requirements, making appropriate software selection, using properly configured data sources, and providing sufficient hardware to meet the user productivity needs
- GIS should be built with performance and scalability to perform during peak operational loads
- System design starts with identifying business needs. This includes identifying user locations and required information products. Identifying required data resources and developing appropriate software applications to do the work.
- A traditional system design process starts with a GIS needs assessment

System Architecture design

- is a process developed by Esri to promote successful GIS enterprise operations
- This process builds on your existing information technology infrastructure and provides specific recommendations for hardware and network solutions based on existing and projected business or user needs
- translates business needs to identify IT requirements.

Distributed Computer Environment

- must be designed to properly support user performance

Building a GIS implementation strategy

- The implementation strategy includes the processes, the requirements phase, the design phase, construction phase, and implementation phase
- During the requirements phase the user information product needs establish a foundation for completing the design
- The design phase includes infrastructure requirements must be identified to quantify deployment costs, Network communication capacity is an important consideration for GIS deployments
- During the construction phase, the construction phase includes system procurement authorization based on the design budget and deployment timeline
- The implementation phase consists of the initial deployment and operational testing, final system delivery, user training, and workflow migration complete. It also includes system maintenance operations

System performance terminology

- A workflow transaction is an average unit of work
- A throughput is a measure of the transaction rate
- Capacity is the maximum rate at which a platform can do work
- Utilization is the percentage of capacity represented by a given throughput rate

System architecture design process

- includes a user needs assessment, a workflow loads analysis, a technical architecture strategy, a user requirements analysis, a network suitability analysis, platform architecture selection, software configuration and enterprise design solution

User needs assessment

- The results of a GIS user needs assessment provides inputs for the system architecture design analysis.

Workflow Loads Analysis

- The workflow loads analysis translates user needs to project workflows with baseline traffic and processing transaction loads based on estimated workflow complexity.

Technical Architecture Strategy

- identifies user locations, network connectivity, and data center server locations.

User Requirements Analysis

- translates peak user workflow loads to peak throughput transaction loads

Network Suitability Analysis

- translates peak site network throughput loads to peak site network traffic and compares with available network bandwidth

Platform architecture selection

- identifies data center platform tier configuration and identity platform selection for each tier

Software Configuration

- identifies platform assignment for each workflow software component with peak transaction processing load

Enterprise Design Solution

- combines all peak workflow software component processing loads on the assigned platform tier, translates baseline processing load to selected platform processing load, and generates the number of nodes required for each platform tier with the estimate of capacity utilization

Batch Process

- is a workflow that does not require user interaction

Tightly scripted software code

- The early ArcInfo software provided developers and professional GIS users with a rich toolkit for geospatial query and analysis and demonstrated the value of GIS technology

Object Relational Software

- Hardware performance improvements led to more efficient programming techniques deployed in the late 1990s

Service Oriented Architecture

- Web technology introduced more ways to share data and services, introducing a services -oriented component architecture along with the interoperability standards that enable open and adaptive applications developed from multi -vendor component architecture

Cloud Computing Platform Architecture

- Hardware Virtualization, Data Center Automation, and Self -Service Cloud Computing provide simpler ways to administer and support GIS applications and services

Distributed Operations

- ArcGIS Enterprise, WebGIS, and Enterprise Solutions provide support for hybrid distributed operations

Four Different Types of GIS

- **Traditional GIS**
    - professionals in GIS departments develop and maintain data sources and create geographic information products from their local work environment
    - GIS users request maps prepared by GIS analysts for use in their work
- **Centralized GIS**
    - In centralized GIS, GIS data is developed and maintained in a centralized geodatabase shared by all departments through the enterprise
    - enterprise application workflows are established with direct user access to dynamic geographic information products used in their work
    - GIS data resides in centralized enterprise geodatabases hosted by relational database management systems.
- **GIS online sharing**
    - shares a selection of common cloud -hosted cached basemaps and imagery that GIS users can share across a wide area network

    - These files can be shared and downloaded to be viewed or analyzed using local GIS desktop cop software
- **Web GIS architecture**
    - ArcGIS Online Organization Services extended GIS user access to directly create and share geospatial data services to users on any device at any location through the cloud
    - Direct access to dynamic geospatial data services, configurable applications for web and mobile phone clients, and real -time geospatial data feeds over a wide area network
    - GIS information products are published as dynamic, REST -based, loosely coupled data services that can be consumed by all GIS applications, desktop, web, or mobile

ArcGIS Enterprise

- The ArcGIS Enterprise deployment concept, managing information from a community of services and delivering web maps to applications supported on every device
- The ArcGIS Portal is a content management -enabled customer engagement, collaboration, and sharing which empowers business users to create and share content.
- Integrated platform architecture facilitates information, development, and sharing.

Capacity Planning Tool, or CPT

- was developed to automate the system design process by Esri
- is an Excel workbook designed as a model to configure business requirements and to generate appropriate platform design solutions
- Microsoft Excel functions are used to automate the design analysis and identify a design solution that satisfies identified business requirements

Information security

- the process of protecting the availability, privacy, and integrity of data. Types of security threats

CIA Security Triad

- The core principles of information security management are represented by the CIA Triad
- includes confidentiality, integrity, and availability
- Confidentiality is protection of privileged communications, restricting user access to core business information based on a need -to -know principle
- Integrity refers to the trustworthiness of business data resources and associated information products generated over its entire lifecycle
- Availability refers to ensuring the information system is functional when needed to support operational business requirements

Security Needs

- More security will result in a higher level of security investment, higher increases in complexity and system costs, restricts access to data sources and online services, eliminates external online collaboration, prevents most connected mobile applications and provides optimum protection to manage security risks
- Basic security requires minimum level of security investment, enables the simplest and lowest system cost, enables full access to internet data sources and online services,

provides optimum business environment for external collaboration, extends enterprise operations to include connected mobile applications and protects system threats

Virtualization

- is the act of creating a virtual rather than actual version of something at the same extraction level, including virtual computer hardware platforms, storage devices, and computer network resources
- Hardware virtualization or platform virtualization refers to the creation of a virtual machine that acts like a real computer with an operating system
- Software executed on these virtual machines is separated from the underlying hardware resources

902 Knowledge of systems and application security

Application security or AppSec

- consists of making applications more secure by finding, fixing, and enhancing the security
- Continuous deployment and integration cause security fixes to be deployed constantly, daily or hourly
- Its final goal is to improve security practices and to find, fix, and preferably prevent security issues within applications
- It encompasses the whole application lifecycle from requirements analysis, design, implementation, verification, as well as maintenance

Security testing

- Static testing analyzes code at fixed points
- Dynamic testing analyzes running code and can simulate an attack
- Interactive testing combines static and dynamic testing
- Mobile testing is designed to examine how an attacker can attack mobile apps

Methods to prevent unauthorized access to data and metadata

- include define who has access, Use other software to enforce these policies
- Identify management systems to check the credentials of users
- And data authenticity verification

Security Threats

- include broken access control, cryptographic failures, injection, insecure design, security misconfiguration, vulnerable and outdated components, identification and authentication failures, software and data integrity failures, security logging and monitoring failures, and server-side request forgery.

903 Awareness of trends in geospatial technology

GIS Upcoming Trends

- Current and upcoming trends in geospatial technology include artificial intelligence in GIS data creation, machine learning in GIS, IoT the Internet of Things, drones and and mobile field data collection

The Internet of Things IOT

- The Internet of Things describes devices with sensors, processing ability, software, and other technologies that connect and exchange data with other devices and systems over their internet or other communications networks
- Cisco Systems estimated that the Internet of Things was born between 2008 and 2009 when the things people ratio going from 0.08 in 2003 to 1.84 in 2010

Five tech trends driving new geospatial development

- One, the real -time revolution. Although real -time spatial temporal data is now being generated almost ubiquitously and its applications in research and commerce are widespread and rapidly accelerating, the ability to continuously create and interact in real -time with this data is a recent phenomenon
- Two, miniaturization of technologies. The capacity to create small and often inexpensive devices and sensors with wireless connectivity is driving an explosion of the Internet of Things. Miniaturized and lower -cost sensors lead to an increase in what, when, where, and how much data is collected and more importantly the ability to attune the sensor to the specific data collection needed
- Three, the proliferation of new mobile geospatial sensor platforms. The rapid miniaturization of technologies has made it feasible to explore new modalities for sensor distribution, such as small satellites and unmanned aircraft systems that can be rapidly designed and deployed with orbits or flight paths tailored to the mission
- Four, expanding wireless and web networks
- Five, advances in computing capacity for geospatial research. High -performance computing networks and cloud computing services are providing governments and others with conduits through which they can more easily and quickly access and contribute to growing repositories of geospatial data, tools, and services.

Professional Practice

1001 Understanding of appropriate interpretation of work-related policies and procedures

- Each company has different policies and procedures for editing, storing and archiving data and those policies should be understood and followed.

1002 Understanding of ethics related to technical GIS work

Four primary categories of ethics

- obligations to society
- obligations to employers and funders
- obligations to colleagues and the profession
- obligations to individuals in society

GISCI's code of ethics

- is a set of implementing laws of professional practice that seek to express the primary examples of ethical behavior consistent with the code of ethics
- Both the code and the rules govern ethical professional practice standards and violations of each may be brought before the GISCI as an ethics issue
- The GIS professional should not interpret the lack of a specific context or act from the rules of conduct as permission to behave in any particular manner

1003 Knowledge of managing, documenting, and communicating GIS work

- There is a significant amount of GIS data being created and modified on a constant basis. Following procedures to manage, document, and communicate this work is extremely important

1004 Awareness of how GIS is used across other professions

- GIS skills and jobs are vastly varying between professions, companies, and industries. Read other LinkedIn profiles to get a sense of what skills others have in GIS. Attend conferences and webinars. Join a local group and discuss with other GIS professionals what tools they use and what kind of work they perform as well as read blogs and social networking.

1005 Awareness of GIS-related professional organizations and certification

- For individuals in the United States here are a few professional organizations. URISA, GISCI, and the American Association of Geographers, AAG
- There are many types of professional GIS certifications outside of academic GIS certifications. The GIS Certification Institute, GISCI, oversees the GIS Professional GISP Certification. GISCI receives its authority to administer the GISP Certification Program from the following five GIS organizations.
    - The Association of American Geographers, AAG
    - The National States Geographic Information Council, NSGIC
    - The University Consortium of Geographic Information Science, UCGIS
    - The Urban and Regional Information Systems Association, URISA
    - Geospatial Information and Technology Association, GITA
- These five organizations are the members of GISCI. The board of directors is comprised of ten people, two representatives from each organization, whom govern the GISCI.